EGE UNIVERSITY
FACULTY OF ENGINEERING
COMPUTER ENGINEERING DEPARTMENT

INTRODUCTION TO DATABASES
2025–2026 FALL SEMESTER

PROJECT REPORT
(Erasmus Curriculum Database Integration)

DELIVERY DATE
21/12/2025

PREPARED BY GROUP 1
İrem Kurtulmaz
Efe Çiçekdağı
Gizem Redif
Mert Ali Berrak

Table Of Content ……1

1. Introduction and Project Scope……...2
   1.1. Project Objective……2
   1.2. Selected Universities and Departments……...2
2. Individual Curriculum Analyses……2
   2.1. Analysis 1: Yaşar University - Software Engineering (English Program).....2
      2.1.1. Structural Analysis……..2
      2.1.2. Raw Data Tables……4
      2.1.3. Difference From Our Other Universities……6
   2.2. Analysis 2: Hacettepe University - Artificial Intelligence Engineering (English Program).....6
      2.2.1. Structural Analysis……..6
      2.2.2. Raw Data Tables……8
      2.2.3. Difference From Our Other Universities……..10
   2.3. Analysis 3: Çukurova University - Computer Engineering (English Program).....10
      2.3.1. Structural Analysis……10
      2.3.2. Raw Data Tables……11
      2.3.3. Difference From Our Other Universities……..13
3. Requirement Analysis & Enhanced Entity–Relationship (EER) Models……13
   3.1. Yaşar University - Software Engineering……..13
      3.1.1. Requirement Analyse……13
      3.1.2. Enhanced Entity-Relationship(EER) Model……14
   3.2. Hacettepe University - Artificial Intelligence Engineering……..15
      3.2.1. Requirement Analyse……15
      3.2.2. Enhanced Entity-Relationship(EER) Model……17
   3.3. Çukurova University- Computer Engineering……..17
      3.3.1. Requirement Analyse……17
      3.3.2. Enhanced Entity-Relationship(EER) Model……19
4. Integrated Design……..19
   4.1. Schema Integration Methodology……19
   4.2. Application of the Schema Integration Methodology……...22
5. Logical Model……...35
   5.1. Mapping……35
   5.2. Relational Schema……37
6. Physical Implementation and Querying……40
   6.1. DBMS Selection……...40
   6.2. Database Schema Definition (SQL DDL Scripts).....40
   6.3. Sample Data Population (SQL Scripts).....46
   6.4. Triggers……56
   6.5. Check Constraints for Curriculum and Equivalency Rules……..59
   6.6. Data Manipulation Examples (INSERT / UPDATE / DELETE).....59
   6.7. Query Set: Core SELECT Statements……..60
   6.8. Academic Insight Queries (Original SELECT Statements).....62
7. References……63

1. Introduction and Project Scope

1.1 Project Objective

The objective of this project is to design and implement an integrated academic database that unifies the EER schemas of three different universities within the scope of the Erasmus program. The project adopts a systematic, literature-based integration approach to resolve inconsistencies and semantic differences among the schemas. Within this framework, the consistent representation of curricula, courses, elective courses, and academic staff is targeted. By producing a unified global schema, the system aims to support meaningful academic analyses, such as course comparison and cross-university course/curriculum matching, while preserving the semantic integrity of the original schemas.

1.2 Selected Universities and Departments

The universities and corresponding departments selected for the Erasmus+ system within the scope of this project are as follows:

- Yaşar University – Software Engineering (English Program)
- Hacettepe University – Artificial Intelligence Engineering (English Program)
- Cukurova University – Computer Engineering (English Program)

The selection of these universities was primarily driven by the structural and curricular diversity observed in their academic programs.
- Specifically, the selected departments exhibit significant differences in how curriculum components are modeled and offered.
- In some programs, elective courses are managed through course pool structures and may be offered in multiple semesters, while in others, electives are strictly semester-based.
- Additionally, certain departments require laboratory components to be taken as separate courses with distinct course codes, whereas others integrate laboratory hours within the main course structure.
- Differences also exist in terms of prerequisite definitions, instructional roles (such as the presence of assistants and coordinators), and the inclusion of additional course attributes beyond standard workload and credit information.

These variations were deliberately considered in order to capture a wide range of academic practices and to ensure that the proposed Erasmus+ system can effectively support diverse curriculum structures and equivalency scenarios.

2. Individual Curriculum Analyses

This section provides an in-depth analysis of each selected university and department, evaluating them against the established criteria to justify their inclusion in the Erasmus+ framework.

2.1 Analysis 1: Yaşar University - Software Engineering (English Program)

Curriculum Bologna-
https://obs.yasar.edu.tr/oibs/bologna/progCourses.aspx?lang=en&curSunit=76

2.1.1 Structural Analysis

1. How is the program organized?

The program is organized into 8 semesters (4 academic years), following a semester-based system. The first one year focuses on fundamental courses (Math, Physics, Programming), while the last three years focus on specialization .

2. How do prerequisite rules affect course sequencing?

Yes, some courses have prerequisites. In this case, it creates a chain that affects your ability to take certain courses.

3. How might courses match or overlap?

Courses such as Data Structures and Algorithms, Analysis and Design of Algorithms, Database Systems, Automata Theory and Operating System and Networking are common among most Computer Engineering programs.

4. Is there an internship or project course?

Yes. The curriculum includes a 7th-semester internship course titled *Internship* and a two-phase Senior Design Project (I–II) in the 7th and 8th semesters.
These courses are project-based and classified as capstone components.

5. Is there a difference between Online and Onsite courses?

The program is primarily on-site, with lectures and labs conducted in person.
However, some courses may include online components

6. Which course domains exist (core/elective/capstone)?

The program includes three main domains:

- Core Courses
- Elective Courses
- Capstone Courses

7. Total semesters and total number of courses

The program consists of 8 semesters and 49 courses, each contributing to the total 240 ECTS required for graduation.

8. Ratio of compulsory vs elective courses

Approximately 86% of the courses are compulsory, while 14% are elective.

9. Prerequisite relationships

Programming and Problem Solving 1 → Programming and Problem Solving 2
Programming and Problem Solving 1 → Data Structures and Algorithms
Occupational Health And Safety 1 → Internship
These prerequisite paths show vertical knowledge dependency, ensuring conceptual progression.

10. How do these chains affect progress?

Course prerequisites are important for a better understanding of the subjects. If a prerequisite is not fulfilled, the duration of the study may be extended.

11. Is the ECTS distribution balanced?

Each semester carries an average of 29–31 ECTS

12. Program Language

The medium of instruction is English. Turkish and Atatürk's Principles and History Of Revolution courses are taught in Turkish.

13. Language proficiency

Students are expected to have at least a B2 English level (CEFR) to effectively follow the program.

14. System: Exam-heavy vs Project-heavy

The program is exam-oriented.

15. Number and distribution of instructors

The department has approximately 14 academic staff, including professors, associate professors, and research assistants.

16. MÜDEK/ABET

The Yaşar University Software Engineering Department is accredited by MÜDEK

2.1.2 Raw Data Table

COURSE PLAN

CourseCode	Course Name	T+A+L	ECTS	Semester	Prerequisites	Description/Learning Outcomes
SE 1105	PROGRAMMING AND PROBLEM SOLVING I	3+2+0	6	1	None	This course is an introduction to programming course with emphasis on problem solving, programming, algorithm design and introductory programming principles
SE 1108	PROGRAMMING AND PROBLEM SOLVING II	3+2+0	5	2	SE 1105	Classes, Methods, Inheritance, Polymorphism, Interfaces, Exceptions, Simple File Processing
COMP 1202	DISCRETE COMPUTATIONAL STRUCTURES	2+2+0	6	3	None	Set theory, functions, relations, graphs, trees
COMP 3330	AUTOMATA THEORY	3+0+0	6	5	COMP 1202	Finite automata, regular expressions, regular languages, pushdown automata, context-free grammars, context-free languages, Turing machines, recursive languages, recursively enumerable languages, Chomsky hierarchy, decidability.
SE 3317	SOFTWARE DESIGN AND ARCHI	2+2+0	6	5	SE 1108	This course covers the principals behind the software design patterns and architectures design patterns, and their application in

	TECTURE					constructing software components.
SE 2230	DATABASE SYSTEMS	2+2+0	6	4	None	This course covers relational model, entity-relationship model, normalization of relations, basic and advanced SQL using constraints, views and triggers, fine tuning of databases using indexes, and using transactions to provide ACID properties.
UNV ELECT SE1	UNIVERSITY ELECTIVE COURSE SE1	None	5	5	None	Relevant course details are in a different table
ELECT SE A1	DEPARTMEN ELECTIVE COURSE A1	None	6	6	None	Relevant course details are in a different table
ELECT SE A3	DEPARTMENT ELECTIVE COURSE A2	None	6	7	None	Relevant course details are in a different table
ELECT SE B1	DEPARTMENT ELECTIVE COURSE B1	None	6	8	None	Relevant course details are in a different table

UNIVERSITY ELECTIVE COURSE

CourseCode	CourseName	T+A+L	ECTS	Prerequisites	Description/Learning Outcomes
ANE 0301	BASIC VISUAL EFFECT I	2+2+0	5	None	Visual effects and compositing techniques will be demonstrated from class projects. Real life video merging with green screen video and mixing with visual effects will be practiced during the lessons.
SOFL 1611	GERMAN I	4+0+0	4	None	• providing students with the necessary tools to communicate daily with a limited number of vocabulary; • Understanding of basic structures such as words and sentences used in everyday life when spoken clearly and slowly.

UFND 5020	ETHIC CULTURE	2+0+0	2	None	The Ethics Course provides students with the awareness, as well as the psychological and behavioral skills, to proactively strive for a quality, secure and structured lifestyle, in which students can take responsibilty for protecting social, cultural and environmental values.

DEPARTMEN ELECTIVE COURSE A

CourseCode	CourseName	T+A+L	ECTS	Prerequisites	Description/Learning Outcomes
COMP 4312	*THEORY OF COMPUTATION*	3+0+0	6	None	Introduction & basic definitions, concept of abstract automata, finite state machines; push-down automata, Turing Machine, Formal languages, context-free languages, the relationship between automata, languages and grammars. Computability and solvability, some applications

DEPARTMEN ELECTIVE COURSE B

CourseCode	CourseName	T+A+L	ECTS	Prerequisites	Description/Learning Outcomes
EEE 4515	BIOMEDICAL IMAGING	3+0+0	6	None	This course presents the fundamentals of medical imaging with a focus on signals and systems aspect and with up-to-date descriptions of instrumentation. Important imaging modalities are described including projection radiography, x-ray computed tomography, nuclear medicine scintigraphy and emission tomography, ultrasound imaging, and magnetic resonance imaging.

2.1.3 Difference From Our Other Universities

Courses have categories. Some of them are Mathematics and Basic Sciences, Engineering, Science, Social Sciences.

Lab courses are not selected separately. They are automatically assigned when a lab course is selected.

Elective course credits can vary. Therefore, the minimum ECTS credit for the semester in which elective courses must be taken is determined, and students select courses to achieve that minimum total ECTS credit.

2.2 Analysis 2: Hacettepe University - Artificial Intelligence Engineering (English Program)

Cirriculum Bologna - https://bilsis.hacettepe.edu.tr/oibs/bologna/index.aspx?lang=tr&curOp=showPac&curUnit=310&curSunit=765#

2.2.1 Structural Analysis

1. How is the program organized?

The program is organized into 8 semesters (4 academic years), following a semester-based system. The first and second years focus on fundamental courses (Math, Physics, Programming). The first semester also includes a wide range of technical, and non field elective courses. The third and fourth years focus on advanced topics, AI courses and practice labs.

2. How do prerequisite rules affect course sequencing?

In the first semester there are no prerequisites. So students can take all compulsory and elective courses. After the first semester the prerequisite chains begin. These chains might affect your ability to take certain courses.

3. How might courses match or overlap?

Because Hacettepe Artificial Intelligence Engineering primarily consists of computer-based courses, it offers courses similar to those in many software and computer-related departments. In addition to core courses like physics and mathematics, courses like programming, data structures, and algorithms overlap with other departments. However, specialized courses specific to AI engineering may not have a perfect equivalent in other departments. Therefore, care should be taken when matching these courses. Furthermore, because there are many elective courses, interdepartmental compatibility can be achieved through elective courses.

4. Is there an internship or project course?

Yes. The curriculum includes a 5th-semester and 7th-semester internship course titled Internship and a two-phase Senior Design Project (I–II) in the 7th and 8th semesters.
These courses are project-based and classified as capstone components.

5. Is there a difference between Online and Onsite courses?

All classes are on-site, and some of them include labs conducted in person.
There isn't any online course in this program.

6. Which course domains exist (core/elective/capstone)?

The program includes three main domains:

- Core Courses
- Elective Courses
- Capstone Courses

7. Total semesters and total number of courses

The program consists of 8 semesters, 39 compulsory, and 210 elective courses. To graduate, 240 ECTS must be completed along with compulsory courses.
Ratio of compulsory vs elective courses
Approximately 16% of the courses are compulsory, while 84% are elective.

8. Prerequisite relationships

Introduction to Programming I → Introduction to Programming Lab I
Completion of 2nd Grade → Internship I
Completion of 3nd Grade → Internship II
Embedded Systems → Embedded Systems Laboratory
Introduction to Deep Learning → Deep Learning Laboratory
Introduction to Data Mining → Data Mining Laboratory

These prerequisite paths show vertical knowledge dependency, ensuring conceptual progression.

9. How do these chains affect progress?

Course prerequisites are important for a better understanding of the subjects. If a prerequisite is not fulfilled, the duration of the study may be extended.

10. Is the ECTS distribution balanced?

Each semester carries an average of 29–31 ECTS

11. Program Language

All classes are taught in English besides Turkish, Atatürk's Principles and History Of Revolution, and elective language courses. Turkish, Atatürk's Principles and History Of Revolution courses taught in English, while elective language courses are delivered in the respective language.

12. Language proficiency

Students are expected to have at least a B2 English level (CEFR) to effectively follow the program.

13. System: Exam-heavy vs Project-heavy

The program is exam-oriented.

14. Number and distribution of instructors

The department has 2 professors, 2 associate professors, 3 instructors and 5 research assistants.

15. MÜDEK/ABET

The Hacettepe University Artificial Intelligence Engineering isn't accredited by any institution

2.2.2 Raw Data Table

COMPULSORY CLASSES

CourseCode	**CourseName**	**T+A+L**	**ECTS**	**Semester**	**Prerequisites**	**Description/Learning Outcomes**
BBM101	INTRODUCTION TO PROGRAMMING I	3+0+0	6	1	BBM103	To instill basic algorithmic thinking and structural programming logic.
BBM103	INTRODUCTION TO PROGRAMMING LAB I	0+2+0	4	1	BBM101	To reinforce theoretical programming knowledge with practical applications and develop debugging skills.
BBM102	INTRODUCTION TO PROGRAMMING II	3+0+0	8	2	BBM104 & BBM103 & BBM101	The fundamentals of object-oriented programming
BBM104	INTRODUCTION TO PROGRAMMING LAB II	0+2+0	4	2	BBM101 & BBM102 & BBM103	The fundamentals of object-oriented programming.
AIN312	FORMAL LANGUAGES AND AUTOMATA THEORY	3+0+3	6	6	BBM102 & BMM104	Introduction to automata theory.Deterministic and nondeterministic finite automata.Regular languages and regular expressions

AIN211	PRINCIPLES OF ARTIFICIAL INTELLIGENCE	3+2+4	6	4	BBM101 & BBM103	Problem-solving agents, complete search algorithms, heuristic search algorithms, game-based algorithms, constraint satisfaction problems, logical agents, propositional logic and first-order logic, inference and decision making

NONTECHNICAL ELECTIVE COURSES

CourseCode	CourseName	ECTS	Prerequisites	Description/Learning Outcomes
EKO115	GENERAL ECONOMY	4	None	Null

AIN TECHNICAL ELECTIVE

CourseCode	CourseName	T+A+L	ECTS	Prerequisites	Description/Learning Outcomes
AIN411	INTRODUCTION TO BIOONFORMATICS	3+0+0	6	(BBM102) ve (BBM104)	The fundamentals of molecular biology and genetics, the structure and properties of data generated at different levels, DNA and protein sequence analysis, protein 3-D structure analysis, gene and protein functional annotation and prediction, phylogenics and population genetics, gene expression analysis, systems biology, and biological networks. Furthermore, examples of AI-based applications from the literature are included for all these topics.
AIN420	INTRODUCTION TO DEEP LEARNING	3+0+0	6	(BBM102) ve (BM104)	Neural networks, learning in neural networks, feedback, supervised deep learning methods, convolutional neural networks, unsupervised deep learning methods, recursive neural network models, fundamentals of reinforcement deep learning, modeling sequence data, fundamentals of deep adversarial networks.

BBM TECHNICAL ELECTIVE

CourseCode	CourseName	T+A+L	ECTS	Prerequisites	Description/Learning Outcomes
BBM486	DESIGN PATTERN	3+0+0	6	BBM102 & BMM104	To teach software design patterns in an applicable way

AIN TECHNICAL ELECTIVE LAB.

CourseCode	CourseName	T+A+L	ECTS	Prerequisites	Description/Learning Outcomes

AIN422	DEEP LEARNING LABORATORY	0+2+0	4	(BM102) and (BBM104)	Neural networks, learning in neural networks, feedback, supervised deep learning methods, convolutional neural networks, unsupervised deep learning methods, recursive neural network models, fundamentals of reinforcement deep learning, modeling sequence data, fundamentals of deep adversarial networks

2.2.3 Difference From Our Other Universities

There are no assistants or coordinators.
Elective courses can also have their own labs.
There is not delivery mode because all classes are on-site.

2.3 Analysis 3: Çukurova University - Computer Engineering (English Program)

Cirriculum Bologna – https://ebs.cu.edu.tr/Program/DersPlan/555/2025

2.3.1 Structural Analysis

1. How is the program organized?

The program is offered as a 4-year undergraduate degree. It is organized on a semester basis, with 2 semesters per academic year, totaling 8 semesters.

2. How do prerequisite rules affect course sequencing?

There are no prerequisite requirements between courses in this department.

3. How might courses match or overlap?

Courses such as Mathematics I, Algorithms and Programming I, Object-Oriented Programming, and Data Structures are common in most Computer Engineering programs, and this similarity also exists in this curriculum.

4. Is there an internship or project course?

Yes, a summer internship is required for graduation. Additionally, “Graduation Thesis” courses are offered in the 7th and 8th semesters.

5. Is there a difference between Online and Onsite courses?

The program is conducted entirely face-to-face; lectures and laboratories are held on campus.

6. Which course domains exist (core/elective/capstone)?

The program includes three main domains:

- Core Courses
- Elective Courses
- Capstone Courses

7. Total semesters and total number of courses

The program consists of 8 semesters and approximately 48 courses, although the exact number may vary depending on the student’s elective choices. A total of 240 ECTS credits is required for graduation.

8. Ratio of compulsory vs elective courses

Approximately 74% of the courses are compulsory, while 26% are elective.

9. Prerequisite relationships
There are no prerequisite chains between courses.

10. Is the ECTS distribution balanced?
Each semester consists of approximately 30 ECTS credits.

11. Program Language
All compulsory and elective department courses are taught 100% in English. Courses such as Turkish Language, Atatürk's Principles and History of Revolution, as well as non-field elective courses, are taught in Turkish.

13. System: Exam-heavy vs Project-heavy
The program is exam-oriented.

14. Number and distribution of instructors
The department has a total of 20 academic staff members, including professors, associate professors, faculty members, lecturers, and research assistants.

15. MÜDEK/ABET
The Çukurova University Computer Engineering Department is accredited by MÜDEK.

2.3.2 Raw Data Table

COURSE PLAN

Course Code	CourseName	T+A	ECTS	Semester	Prerequisites	Description/Learning Outcomes
CEN111	ALGORITHMS AND PROGRAMMING I	4+0	4	1	None	Problem solving fundamentals. Algorithm Design. Programming Basics. Program control structures. Modular Programming
CEN113	ALGORITHMS AND PROGRAMMING LAB. I	0+2	2	1	None	Implementation of software from CEN111 Algorithms And Programming I course
CEN116	ALGORITHMS AND PROGRAMMIG II	3+0	4	2	None	Structural programming concept. C Programming Language, Fundamentals, Data types, Statements, String functions. Array manipulations. Procedures and Functions. Recursion. Sorting and Searching algorithms. Basic file applications. Dynamic variables and elementary data structures, Pointer, Stack, Queue, Linked list
CEN201	ALGORITHMS AND PROGRAM	0+2	2	2	None	Implementation of software in the CEN116 Algorithms and Programming I course content

	MING LAB. II					
CEN343	THEORY OF COMPUTATION	3+0	5	5	None	Chomsky hierarchy, Regular languages, Context free languages, Turing machines, Decidability, P and NP languages, NP-complete languages
CEN202	PROGRAMMING LANGUAGES	3+0	6	4	None	Basic programming knowledge
CEN301	DATA MANAGEMENT AND FILE STRUCTURES	3+0	6	5	None	An overview of data storage and indexing methods. Disks and files. RAID systems. Disk space management. Record-based files. Tree-based directories (B+ Tree, B Tree, LSM Tree). Hash indexes (static hash, extendible hash, linear hash indexes). Secondary key search (k-dimensional trees, guide files). Bitmap indexes. External sorting. Data abstraction with the entity-relationship model. Introduction to database management system concepts. Introduction to big data concepts.

DEPARTMEN ELECTIVE COURSE

Course Code	CourseName	T+A	ECTS	Semester	Prerequisites	Description/Learning Outcomes
CEN311	LINEAR SYSTEMS	3+0	5	5	None	Fourier series, Fourier transforms, transfer functions of continuous and discrete-time systems, transient and steady-state response, natural response and stability, convolution.
CEN312	COMPUTER GRAPHICS	3+0	5	6	None	Basic computer graphics algorithms are examined by using OpenGL applications. 2D and 3D graphics techniques, modelling of shapes, illumination, shading, texture mapping methods and algorithms are covered.
CEN405	INTRODUCTION TO ROBOTICS	3+0	6	7	None	General concepts, Transformations, Advanced kinematics, Inverse kinematics, Jacobian, Orbit Planning

NON-FIELD ELECTIVE COURSE

Course Code	CourseName	T+A	ECTS	Term	Prerequisites	Description/Learning Outcomes

SD0367	EXAMINAI İONS ON MODERN TURKISH LITERATU RE I	2+0	3	Fall	None	The content of this course is modern Turkish poetry in the context of poetry analysis related to Tanzimat, Servet-i Funun, National Literature and Turkish poetry in the Republican period.
SD0499	IMPORTA NCE OF INDUSTRI AL RAW MATERIA LS IN OUR LIFE	2+0	3	Fall and Spring	None	Introduction of mineral deposits formed by the effect of external events, Effects on mineral deposits in geological and chemical events, knowledge of alterations, Explanation of these mineral deposits.
SD0534	CREATIVE WRITING	2+0	3	Spring	None	Personal essay, academic essay, fiction, free writing, grammar, writing game, poem.

2.3.3 Difference From Our Other Universities

Courses do not have prerequisites. To take any course, students are not required to fulfill any conditions related to other courses.

There is no fixed pool for departmental elective courses; students select from the elective courses offered in that semester to meet the required ECTS credits.

Departmental elective courses and non-departmental elective courses do not include laboratory components.

3. Requirement Analysis & Enhanced Entity–Relationship (EER) Models

3.1 Yaşar University - Software Engineering

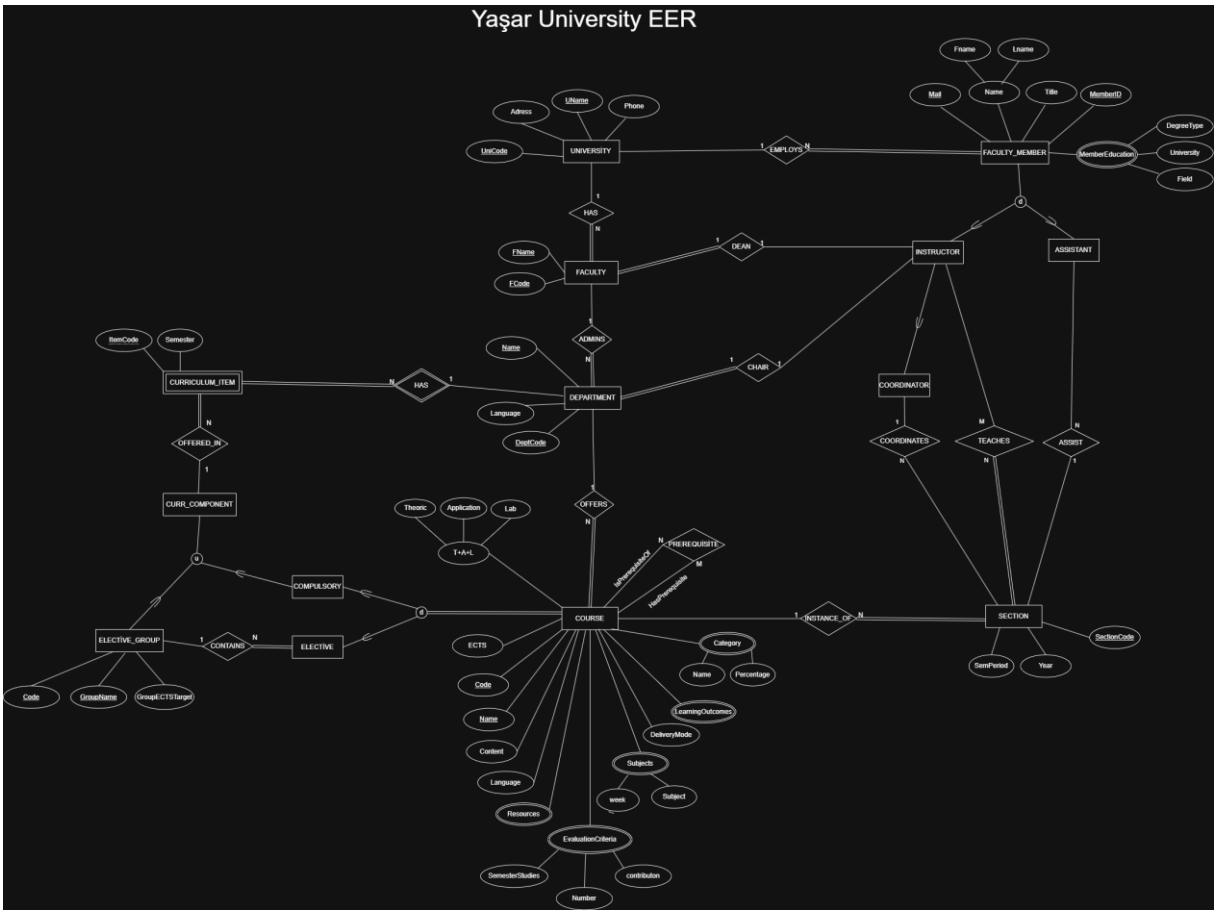
3.1.1 Requirement Analysis

The system manages information about universities, faculties, departments, faculty members, courses, sections, and curriculum structures.

- Each UNIVERSITY has a unique UniCode, a UName, an Address, and a Phone.
- Each UNIVERSITY may have zero or more FACULTY entities.
- Each FACULTY has a unique FCode and a unique FName.
- Each FACULTY must belong to exactly one UNIVERSITY.
- Each FACULTY administers zero or more DEPARTMENT entities.
- Each DEPARTMENT has a unique DeptCode, a unique Name, and a Language.
- Each DEPARTMENT must be administered by exactly one FACULTY.
- Each COURSE has a unique Code, a Name, Content, Language, Resources, DeliveryMode and ECTS.
- A COURSE can have more than one resources.
- Each COURSE must be offered by exactly one DEPARTMENT.
- A DEPARTMENT may offer zero or more COURSE entities.
- Each COURSE includes workload information consisting of Theoric, Application, and Lab, represented as the composite attribute T+A+L.
- Each COURSE may have one or more Category values, each defined by Name and Percentage.
- Each COURSE may cover one or more Subjects, each defined by Subject and Week.
- Each COURSE may have one or more LearningOutcomes.
- Each COURSE may include one or more EvaluationCriteria, each consisting of Number, Contribution, and SemesterStudies.

- A COURSE may have zero or more prerequisite courses.
- The PREREQUISITE relationship between courses is many-to-many and is represented by the roles isPrerequisiteOf and hasPrerequisite.
- A SECTION must be an instance of exactly one COURSE.
- Each SECTION has a unique SectionCode, a SemesterPeriod, and a Year.
- A COURSE may have zero or more SECTION entities.
- Each FACULTY_MEMBER has a unique MemberID, a unique Mail , a Name (composed of FName and LName), and a Title.
- Each FACULTY_MEMBER must be employed by exactly one UNIVERSITY.
- Each UNIVERSITY may have zero or more FACULTY_MEMBER entities.
- A FACULTY_MEMBER may have one or more MemberEducation records, each including a DegreeType, a University, and a Field.
- FACULTY_MEMBER is specialized into INSTRUCTOR and ASSISTANT through a disjoint specialization.
- A FACULTY_MEMBER may belong to at most one specialization subtype.
- Each FACULTY must have exactly one DEAN, who is an INSTRUCTOR.
- Each DEPARTMENT must have exactly one CHAIR, who is an INSTRUCTOR.
- An INSTRUCTOR may serve as the DEAN of a FACULTY.
- An INSTRUCTOR may serve as the CHAIR of a FACULTY.
- An INSTRUCTOR may serve as a COORDINATOR.
- A COORDINATOR may coordinate zero or more SECTION entities.
- Each SECTION may have at most one COORDINATOR.
- An INSTRUCTOR may teach zero or more SECTION entities.
- Each SECTION must be taught by at least one INSTRUCTOR.
- An ASSISTANT may assist in at most one SECTION entities.
- Each SECTION may be assisted by zero or more ASSISTANT entities.
- COURSE is specialized into COMPULSORY and ELECTIVE through a disjoint specialization.
- Each COURSE must belong to exactly one specialization subtype.
- Each ELECTIVE_GROUP has a unique Code, a GroupName, and a GroupECTSTarget.
- Each ELECTIVE_GROUP may contain zero or more ELECTIVE courses.
- Each ELECTIVE course must belong to exactly one ELECTIVE_GROUP.
- CURR_COMPONENT is defined as a category formed by the union of COMPULSORY and ELECTIVE_GROUP entities.
- A CURRICULUM_ITEM can’t exist without a DEPARTMENT entity.
- Each CURRICULUM_ITEM has an ItemCode and a Semester, and an ItemCode may be reused across different curricula
- Each CURRICULUM_ITEM must be associated with exactly one DEPARTMENT.
- Each DEPARTMENT may have zero or more CURRICULUM_ITEM.
- Each CURRICULUM_ITEM must offer exactly one CURR_COMPONENT.
- A CURR_COMPONENT may be offered in zero or more CURRICULUM_ITEM entities.

3.1.2 Enhanced Entity-Relationship(EER) Model



3.2 Hacettepe University - Artificial Intelligence Engineering

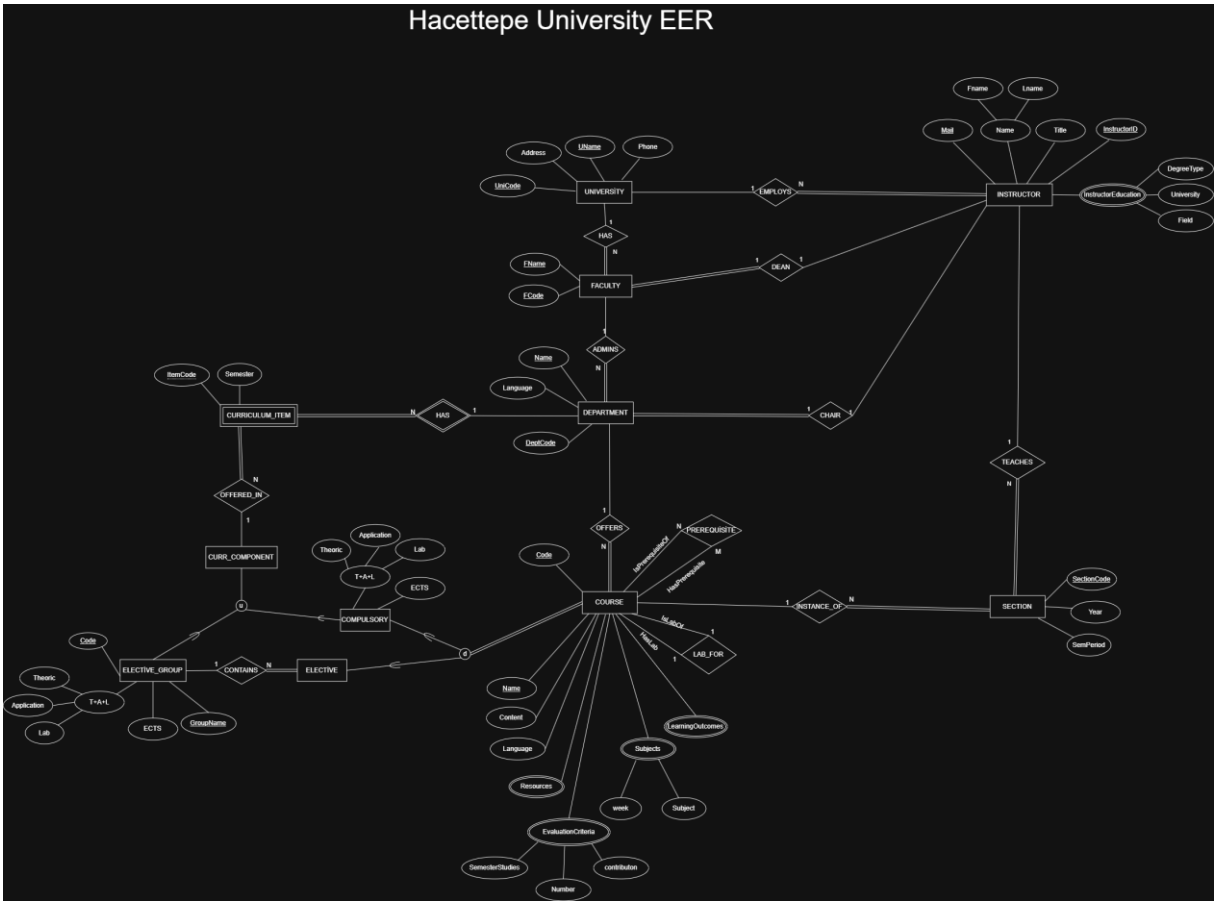
3.2.1 Requirement Analysis

The system manages information about universities, faculties, departments, faculty members, courses, sections, and curriculum structures.

- Each UNIVERSITY has a unique UniCode, a unique UName, an Address, and a Phone.
- Each UNIVERSITY may employ zero or more INSTRUCTOR entities.
- Each UNIVERSITY may have zero or more FACULTY entities.
- Each FACULTY has a unique FCode and a unique FName.
- Each FACULTY must belong to exactly one UNIVERSITY.
- Each FACULTY administers zero or more DEPARTMENT entities.
- Each DEPARTMENT has a unique DeptCode, a unique Name, and a Language.
- Each DEPARTMENT must be administered by exactly one FACULTY.
- Each INSTRUCTOR has a unique InstructorID, a Name (composed of FName and LName), a unique Email, and a Title.
- Each INSTRUCTOR must be employed by exactly one UNIVERSITY.
- An INSTRUCTOR may have zero or more INSTRUCTOR_EDUCATION records, each having a DegreeType, a University, and a Field.
- Each FACULTY must have exactly one DEAN, who is an INSTRUCTOR.
- Each DEPARTMENT must have exactly one CHAIR, who is an INSTRUCTOR.
- An INSTRUCTOR may serve as the DEAN of a FACULTY.
- An INSTRUCTOR may serve as the CHAIR of a DEPARTMENT.
- Each COURSE has a unique Code, a unique Name, Content, Language and Resources.
- Each COURSE must be offered by exactly one DEPARTMENT.
- A DEPARTMENT may offer zero or more COURSE entities.
- Each COURSE may cover one or more Subjects, each defined by Subject and Week.
- Each COURSE may have one or more LearningOutcomes.

- Each COURSE may include one or more EvaluationCriteria, each consisting of Number, Contribution, and SemesterStudies.
- A COURSE may have zero or more prerequisite courses.
- The PREREQUISITE relationship between courses is many-to-many and is represented by the roles isPrerequisiteOf and hasPrerequisite.
- A COURSE may have at most one COURSE acting as a lab through the LAB_FOR relationship (hasLab).
- A COURSE acting as a lab must be related to exactly one main COURSE through the LAB_FOR relationship (isLabOf).
- A SECTION must be an instance of exactly one COURSE.
- Each SECTION has a unique SectionCode, a SemesterPeriod, and a Year.
- A COURSE may have zero or more SECTION entities.
- Each SECTION must be taught by exactly one INSTRUCTOR.
- An INSTRUCTOR may teach zero or more SECTION entities.
- Each COMPULSORY component has ECTS.
- Each COMPULSORY component includes workload information consisting of Theoric, Application, and Lab, represented as the composite attribute T+A+L.
- COURSE is specialized into COMPULSORY and ELECTIVE through a disjoint specialization.
- Each COURSE must belong to exactly one specialization subtype.
- Each ELECTIVE_GROUP has a unique Code and a unique GroupName, ECTS and workload information consisting of Theoric, Application, and Lab, represented as the composite attribute T+A+L.
- Each ELECTIVE_GROUP may contain zero or more ELECTIVE components.
- Each ELECTIVE component must belong to exactly one ELECTIVE_GROUP.
- CURR_COMPONENT is defined as a category formed by the union of COMPULSORY and ELECTIVE_GROUP entities.
- A CURRICULUM_ITEM can’t exist without a DEPARTMENT
- Each CURRICULUM_ITEM has a unique ItemCode and a Semester.
- Each CURRICULUM_ITEM must be associated with exactly one DEPARTMENT.
- Each DEPARTMENT may have zero or more CURRICULUM_ITEM.
- Each CURRICULUM_ITEM must offer exactly one CURR_COMPONENT.
- A CURR_COMPONENT may be offered in zero or more CURRICULUM_ITEM entities.

3.2.2 Enhanced Entity-Relationship(EER) Model



3.3 Çukurova University- Computer Engineering

3.3.1 Requirement Analysis

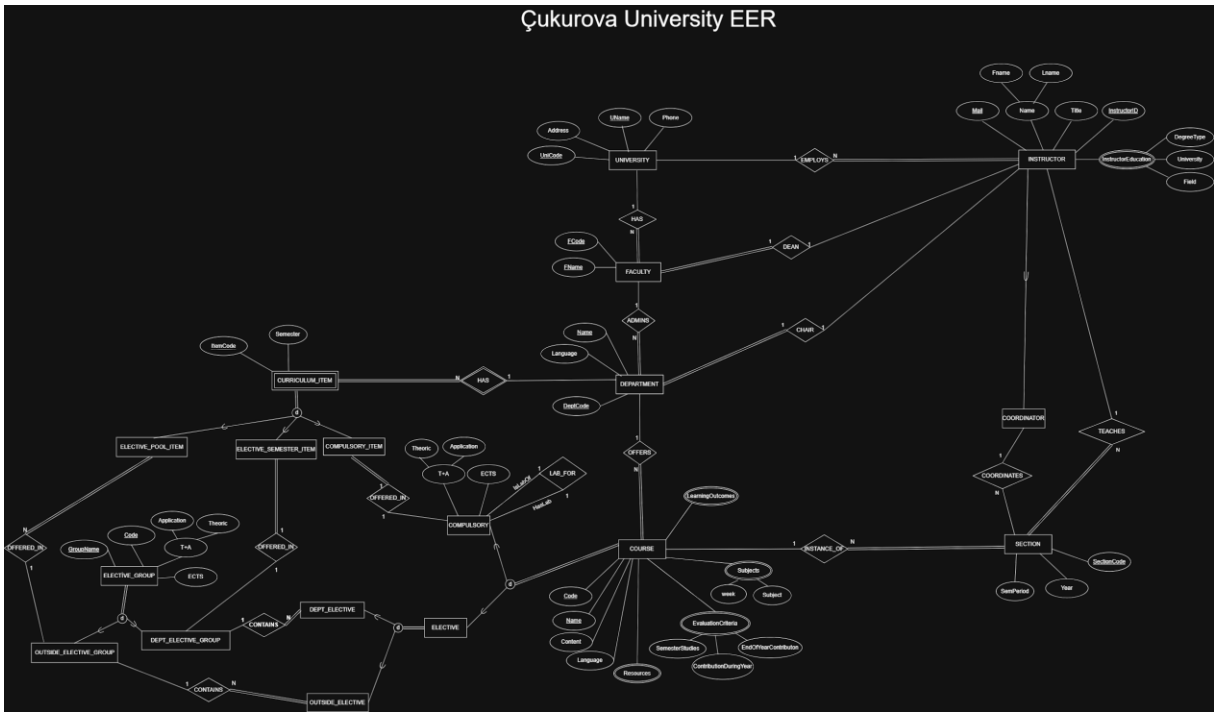
The system manages information about universities, faculties, departments, faculty members, courses, sections, and curriculum structures.

- Each UNIVERSITY has a unique UniCode, a unique UName, an address, and a phone number.
- Each UNIVERSITY may employ zero or more INSTRUCTOR entities.
- Each UNIVERSITY may have zero or more FACULTY entities.
- Each FACULTY has a unique FCode and a unique FName.
- Each FACULTY must belong to exactly one UNIVERSITY.
- Each FACULTY administers zero or more DEPARTMENT entities.
- Each DEPARTMENT has a unique DeptCode, a unique Name, and a Language.
- Each DEPARTMENT must be administered by exactly one FACULTY.
- Each INSTRUCTOR has a unique InstructorID, a Name (composite: FName, LName), a unique Email, and a Title.
- Each INSTRUCTOR must be employed by exactly one UNIVERSITY.
- An INSTRUCTOR may have multiple InstructorEducation records, each consisting of DegreeType, University, and Field.
- Each FACULTY must have exactly one DEAN, who is an INSTRUCTOR.

- Each DEPARTMENT must have exactly one CHAIR, who is an INSTRUCTOR.
- An INSTRUCTOR may serve as the DEAN of a FACULTY.
- An INSTRUCTOR may serve as the CHAIR of a DEPARTMENT.
- Each COURSE has a unique Code, a unique Name, Content, Language and Resources.
- Each COURSE must be offered by exactly one DEPARTMENT.
- A DEPARTMENT may offer zero or more COURSE entities.
- Each COURSE may have one or more LearningOutcomes
- Each COURSE may cover one or more Subjects, each defined by Subject and Week.
- Each COURSE may include one or more EvaluationCriteria, each consisting of EndOfYearContribution, ContributionDuringYear, and SemesterStudies.
- A SECTION must be an instance of exactly one COURSE.
- Each SECTION has a unique SectionCode, a SemesterPeriod and a Year.
- A COURSE may have zero or more SECTION entities.
- Each SECTION must be taught by exactly one INSTRUCTOR.
- An INSTRUCTOR may teach zero or more SECTION entities.
- An INSTRUCTOR may serve as a COORDINATOR.
- A COORDINATOR may coordinate zero or more SECTION entities.
- Each SECTION may be coordinated by at most one COORDINATOR.
- COURSE is specialized into COMPULSORY and ELECTIVE through a disjoint specialization.
- Each COURSE must belong to exactly one specialization subtype.
- Each COMPULSORY course has ECTS and workload information through T+A (composed of theoretical and application components).
- A COMPULSORY course may have at most one COMPULSORY course acting as a lab through the LAB_FOR relationship (hasLab).
- A COMPULSORY course acting as a lab must be related to exactly one main COMPULSORY course through the LAB_FOR relationship (isLabOf).
- ELECTIVE is specialized into DEPT_ELECTIVE and OUTSIDE_ELECTIVE through a disjoint specialization.
- Each ELECTIVE must belong to exactly one specialization subtype.
- ELECTIVE_GROUP specialized into OUTSIDE_ELECTIVE_GROUP and DEPT_ELECTIVE_GROUP.
- Each ELECTIVE_GROUP must belong to exactly one specialization subtype
- Each ELECTIVE_GROUP has a unique Code, GroupName, ECTS and workload information through T+A (composed of theoretical and application components).
- Each OUTSIDE_ELECTIVE_GROUP may contain zero or more OUTSIDE_ELECTIVE entities.
- Each OUTSIDE_ELECTIVE must belong to exactly one OUTSIDE_ELECTIVE_GROUP.
- Each DEPT_ELECTIVE_GROUP may contain zero or more DEPT_ELECTIVE entities.
- Each DEPT_ELECTIVE must belong to exactly one DEPT_ELECTIVE_GROUP.
- A CURRICULUM_ITEM can’t exist without a DEPARTMENT.
- Each CURRICULUM_ITEM has a unique ItemCode and a Semester.
- Each CURRICULUM_ITEM must be associated with exactly one DEPARTMENT.
- Each DEPARTMENT may have zero or more CURRICULUM_ITEM.
- CURRICULUM_ITEM is specialized into COMPULSORY_ITEM, ELECTIVE_SEMESTER_ITEM and ELECTIV_POOL_ITEM through a disjoint specialization.
- Each CURRICULUM_ITEM must belong to exactly one specialization subtype.
- Each ELECTIVE_POOL_ITEM must offer exactly one OUTSIDE_ELECTIVE_GROUP.
- Each OUTSIDE_ELECTIVE_GROUP may be offered in zero or more ELECTIVE_POOL_ITEM entities.
- Each ELECTIVE_SEMESTER_ITEM must offer exactly one DEPT_ELECTIVE_GROUP.
- Each DEPT_ELECTIVE_GROUP may be offered in zero or one ELECTIVE_SEMESTER_ITEM entities.
- Each COMPULSORY_ITEM must offer exactly one COMPULSORY.

- Each COMPULSORY may be offered in at most one COMPULSORY_ITEM entities.

3.3.2 Enhanced Entity-Relationship(EER) Model



4. Integrated Design

4.1 Schema Integration Methodology

Schema integration can be examined in two contexts: View Integration and Database Integration. In the Erasmus project, the objective is to merge EER diagrams belonging to different departments into a single integrated schema.Therefore, the method used falls within the View Integration category. This study is based on the processes proposed by Batini and Lenzerini (1984) and Batini, Lenzerini, and Navathe (1986), which systematically examine database integration methods.

4.1.1 Preintegration

At this stage, the schemas to be integrated and the order of integration are determined.In the literature, four main strategies are proposed for schema integration (Batini et al., 1986):

- Binary–Ladder Strategy: This strategy is based on integrating two schemas at each step. Due to its low complexity, it was preferred for this project.
- Binary–Balanced Strategy
- n-ary One-Shot Strategy
- n-ary Iterative Strategy

In addition, Batini & Lenzerini (1984) state that a weight (schema weight) can be assigned to each schema during the integration process, and that schemas considered more reliable or institutionally more critical should be given priority when resolving conflicts.

4.1.2 Comparison of Schemas

At this stage, the common concepts, differences, and conflicts between the schemas are identified (Batini et al., 1986).

4.1.2.1 Naming Conflicts

Homonym: The same name represents different concepts in different schemas.
Synonym: Different names represent the same concept in different schemas.

Batini & Lenzerini (1984) address these types of naming problems within a more general framework by defining the concept of interschema multiname anomaly and it is an indication that may correspond to synonymy or homonymy depending on context. This concept is a higher-level type of naming conflict that encompasses both homonym and synonym cases and, conceptually, represents the same problem as the naming conflicts defined by Batini et al. (1986).

4.1.2.2 Structural Conflicts

Structural conflicts refer to inconsistencies arising from the modeling approaches of concepts within schemas. Such conflicts are examined in the literature under the following categories (Batini et al., 1986):

- Type Conflict: The same concept is modeled as an attribute in one schema and as an entity or relationship in another schema.
- Dependency Conflict: Differences in the definition of cardinality, participation constraints, or functional dependencies across schemas.
- Key Conflict: The use of different key attributes for the same concept or inconsistencies in key definitions.
- Behavioral Conflict: Differences in behaviors defined during insert/delete operations; for example, whether related entities are deleted when an entity is removed.

Batini & Lenzerini (1984) define two complementary concepts for evaluating such structural differences:

- Concept Likeness: Concepts that have similar modeling characteristics despite having different names
- Concept Unlikeness: Concept unlikeness refers to concepts sharing the same name but exhibiting incompatible modeling characteristics.

This conceptual distinction enables a more accurate identification of potential synonym and homonym relationships in the analysis of structural conflicts and supports the schema comparison process.

4.1.2.3 Other Modeling Differences Identified During Schema Comparison

In addition to naming and structural conflicts, some differences reflecting modeling granularity were also identified.

4.1.3 Conforming

The conforming stage aims to resolve the conflicts identified during the comparison process and to make the schemas ready for integration. This stage consists of three main steps.

4.1.3.1 Name Conflict Resolution

According to Batini and Lenzerini (1984), the resolution of naming conflicts is achieved by properly changing and standardizing the names of the involved objects. In this process, a single consistent name is selected for each concept in the integrated schema, and all related entities, attributes, and relationships are renamed accordingly. As a result, terminological consistency is ensured before proceeding to structural transformations.

4.1.3.2 Syntactic Restructuring:

In this step, the identified conflicts are resolved using appropriate schema transformations (Batini & Lenzerini, 1984). The objective is to first transform elements (entity, attribute, relationship) that represent the same concept but are modeled using different types into the same

modeling type. Within this scope, the transformations defined in the literature can be applied:

T1: Attribute → Entity

T2: Entity → Attribute

Through these transformations, structural consistency between schemas is achieved, and a common representation is created for the next stage.

4.1.3.3 Compatibility Checking:

After syntactic restructuring, the compatibility between concepts is evaluated. The study addresses this evaluation under three categories:

- Identical: All modeling characteristics (keys, cardinalities, constraints) are exactly the same.In this case, the concepts are directly superimposed.
- Compatible: The concepts are not completely identical, but there are no contradictions between them.In this case, the concepts are also considered superimposable.
- Incompatible:

  The following types of conflicts exist between the concepts:

  - Different cardinality definitions for the same concept
  - An attribute defined as an identifier in one schema but not defined as such in the other
  - Inconsistencies in subset/generalization direction
  - Conflicting functional dependencies

  Two main approaches can be applied to resolve incompatible cases:

  Solution 1: Representation Selection One schema is considered more reliable or more comprehensive than the other and is selected as the dominant representation.

  Solution 2: Common Model Creation

  - A new common model that supports the constraints of both schemas is defined.
  - In this case, one concept is generally modeled as a subset or specialization of the other.

Such decisions involve conceptual ambiguity and are therefore made based on user and domain expert consultations

4.1.3.4 Resolution of Compatible but Non-Identical Representations.

According to the compatibility checking criteria proposed by Batini and Lenzerini (1984), several differences identified during the schema comparison phase were classified as compatible but non-identical representations. These cases do not indicate conceptual incompatibility; rather, they arise from different levels of modeling granularity adopted in the schemas. Since no semantic contradictions exist between the representations, these differences were resolved during the conforming stage by extending the less detailed schema. Missing attributes and modeling elements were incorporated to preserve semantic completeness and ensure consistency in the integrated schema.

4.1.4 Merging & Restructuring

At this stage, the conforming schemas are merged using the superposition principle. After the merging process, the resulting global schema is analyzed according to the evaluation criteria proposed in the literature (Batini et al., 1986).

4.1.4.1 Completeness & Correctness

This criterion aims to evaluate whether the semantic information contained in all subschemas is preserved as a result of the merging process. Within this scope, the following questions are addressed:

- Are all concepts and constraints from the schemas correctly represented in the global schema?
- Has any information loss, incorrect generalization, or incomplete representation occurred during the merging process?

4.1.4.2 Minimality

The minimality criterion examines whether the same concept is unnecessarily represented more than once in the global schema. At this stage: Semantically equivalent redundancies are identified, Unnecessary repetitions are eliminated to simplify the schema

4.1.4.3 Understandability

This criterion evaluates the readability, comprehensibility, and maintainability of the resulting EER diagram. When necessary, rearrangements may be made to simplify the overall structure of the global schema. Batini & Lenzerini (1984) additionally recommend performing certain technical analyses during the restructuring process following the merging stage.

4.1.4.4 Redundant Cycles Analysis

After integration, cycles may occur among the relationships in the schema. These cycles are classified as follows:

- Cycles that cannot be removed without information loss
- Cycles that lead to redundant information
- Cycles that introduce unnecessary complexity into the schema

It is recommended to identify such cycles and perform simplification whenever possible.

4.1.4.5 Normalization After Integration

After merging, new functional dependencies may emerge among attributes originating from different schemas. In such cases, the creation of new entities and normalization of the schema are recommended (Batini & Lenzerini, 1984).

4.2 Application of the Schema Integration Methodology

Schema integration was carried out using the methods proposed in the literature within the framework of the integration process to be followed.

Since the integration problem addressed in this project aims to combine different schemas under a single global schema, the view integration approach was adopted.

The integration process was initiated with the preintegration stage.

4.2.1 Preintegration

In this project, three schemas were integrated:
- Yasar University Software Engineering EER Schema
- Hacettepe University Artificial Intelligence Engineering EER Schema
- Cukurova University Computer Engineering EER Schema

While determining the integration order, the Binary–Ladder Strategy proposed in the literature was

used.
According to this strategy, schemas are merged in pairwise stages, and a Partial Integrated Schema (PIS) is obtained as a result of each merging step.

Within the scope of the Binary–Ladder Strategy, the Yasar University Software Engineering EER schema, which was considered to have a more complex structure, was selected as the first schema in the integration process.
The main reasons for prioritizing this schema are as follows:

- The Faculty Member structure includes more complex and multi-level hierarchical relationships.
- The representation of laboratory courses differs from that of the other schemas.

These characteristics made it appropriate to consider this schema as the reference schema in the integration process.

Accordingly, the integration process was carried out in two stages:
Stage 1:
Yasar University Software Engineering EER – Hacettepe University Artificial Intelligence Engineering EER

As a result of merging these two schemas, the first Partial Integrated Schema (PIS) was obtained.

Stage 2:
PIS – Cukurova University Computer Engineering EER As a result of this merging, the final integrated EER schema (Global Integrated Schema) was created.

Stage 1: Yasar University Software Engineering EER - Hacettepe University Artificial Intelligence Engineering EER
At this stage, comparison, conforming, and merging steps were applied between the two schemas, and the resulting outcome was defined as a partial integrated schema (PIS).

4.2.2 Comparison of Schemas

As a result of the comparison conducted between the two schemas, a total of nine differences were identified. These differences are examined in detail under the groups specified below. Apart from these, no structural or semantic differences were observed in the remaining components.

4.2.2.1 Naming Conflicts

- InstructorEducation attribute in Hacattepe University's diagram has the same purpose as the MemberEducation attribute in Yasar University's diagram.
- InstructorId attribute in Hacattepe University's diagram has the same purpose as the MemberId attribute in the Yasar University's diagram.

These are the synonymous naming conflicts between two diagrams.

4.2.2.2 Structural Conflicts

- Type Conflict Analysis:

No cases were encountered in which the same concepts were modeled using different types (entity, attribute, or relationship) in the two schemas. Therefore, no additional conforming was required within the scope of type conflicts.

- Dependency Conflict Analysis:

- In the Yasar University schema, the INSTRUCTOR entity is modeled as a subclass of the FACULTY_MEMBER entity. In the Hacettepe University schema, however, the INSTRUCTOR entity is located at the top level of the hierarchy. For this reason, the EMPLOYS relationship, which is associated with the FACULTY_MEMBER entity in the Yasar schema, is directly associated with the INSTRUCTOR entity in the Hacettepe schema.

- T+A+L and ECTS attributes:
  In the YasarUniversity schema, the T+A+L and ECTS attributes are directly associated with the COURSE entity, whereas in the Hacettepe University schema, these attributes are linked to the ELECTIVE_GROUP and COMPULSORY entities in line with the principle of reducing data redundancy. This situation constitutes a dependency conflict resulting from assigning the functional dependencies of these attributes to different keys in the two schemas.

- Key Conflict Analysis:
  As a result of comparing the primary key definitions, no incompatibility at the key level was identified between the two schemas. Therefore, no additional key conforming operation was required at this stage.

- Behavioral Conflict Analysis:
  No differences were observed between the two schemas in terms of insert/delete behaviors. Accordingly, no behavioral conflict was identified.

- Concept Unlikeness
  At this stage, cases were identified in which the same or similar concepts were incompatible in terms of their modeling characteristics.

  Instructor hierarchy (Subclass structure): In the Yasar University schema, the FACULTY_MEMBER entity is divided into the INSTRUCTOR and ASSISTANT subclasses; additionally, a COORDINATOR subclass is defined under the INSTRUCTOR entity. In contrast, no such hierarchical distinction exists in the Hacettepe University schema. This situation was evaluated as a concept unlikeness arising from the different modeling of dependency structures related to the instructor concept in the two schemas.

- Other Modeling Differences Identified During Schema Comparison

  The Category attribute present in the COURSE entity in the Yasar University schema does not have a counterpart in the Hacettepe University schema.

  The DeliveryMode attribute present in the COURSE entity in the Yasar University schema is not included in the Hacettepe University schema.

  The GroupECTSTarget attribute present in the ELECTIVE_GROUP entity in the Yasar University schema has no corresponding attribute in the Hacettepe University schema.

  The LAB_FOR relationship present in the COURSE entity in the Hacettepe University schema was identified as a relationship that has no counterpart in the Yasar University schema.

4.2.3 Conforming

At this stage, the structural and semantic conflicts identified during the comparison process were resolved using the conforming methods proposed in the literature.The applied solutions are presented below under subheadings.

- Naming Conflict Resolutions:

  The name of the InsEducation attribute in Hacettepe University's diagram is changed to Member_Education.

  The name of the InstructorId attribute in Hacettepe University's diagram is changed to MemberId.

- Resolution of the T+A+L / ECTS Dependency Conflict:

  In the Yasar University schema, the T+A+L and ECTS attributes are modeled on an individual course basis, directly dependent on the key of the COURSE entity. Since this approach provides higher semantic integrity and a greater level of detail at the course level, it was adopted as the reference model for the global schema. Accordingly, attribute-level restructuring was applied to the Hacettepe University schema for conforming purposes. The T+A+L and ECTS attributes located in the COMPULSORY and ELECTIVE_GROUP entities in the Hacettepe schema were removed from these entities and transferred as attributes to the corresponding COURSE entities.

  As a result of this transformation:

  - Group-based redundancies were eliminated,
  - The functional dependency of the attributes was standardized to the COURSE entity in the global schema,
  - The dependency conflict between the schemas was resolved.

- Resolution of the Instructor Subclass Hierarchy (Concept Unlikeness)

  In terms of the hierarchical structure related to the instructor concept, it was determined that the Yasar University schema provides a richer and more detailed model. In accordance with the completeness principle, this structure was preserved for the global schema.

  For this reason, syntactic restructuring was applied to the Hacettepe University schema, and the following modifications were made:

- The FACULTY_MEMBER superclass was added to the Hacettepe schema.
- The existing INSTRUCTOR entity was moved under this superclass.
- The ASSISTANT subclass was added in PIS as a disjoint specialization alongside INSTRUCTOR under the FACULTY_MEMBER entity.
- The COORDINATOR subclass was added under the INSTRUCTOR entity.
- The ASSISTANT entity was connected to the SECTION entity via the ASSIST relationship.
- Roles derived from INSTRUCTOR in the Hacettepe schema (e.g., DEAN, CHAIR) were preserved as INSTRUCTOR subclasses in accordance with the Yasar model.

  With these modifications, the instructor concept in the two schemas was made hierarchically consistent and integrated.

- Resolution of the EMPLOYS Relationships (Dependency Conflict)

  During the compatibility checking step, Solution 1: Representation Selection was applied in order to resolve this incompatibility. Within this scope, based on the structure obtained in the previous step, the FACULTY_MEMBER entity, which is located at the upper level of the hierarchy in the Hacettepe University EER schema, was selected as the dominant representation.

  Accordingly, in the Hacettepe University EER schema, the EMPLOYS relationship was directly associated with the FACULTY_MEMBER entity. As a result, employment relationships were modeled in a consistent and unified manner, and the schemas were made compatible for the subsequent stage of integration.

- Addition of Missing Attributes and Relationships

All concept unlikeness cases identified during the comparison process were mutually resolved in order to ensure the completeness of the global schema.

Within this scope:
Components added from the Yasar schema to the Hacettepe schema:
- GroupECTSTarget attribute → added to the ELECTIVE_GROUP entity
- Category attribute → added to the COURSE entity
- DeliveryMode attribute → added to the COURSE entity

Components added from the Hacettepe schema to the Yasar schema:
- LAB_FOR relationship → defined between COURSE – COURSE

With these additions, both schemas were made semantically complete and compatible.

After completing all the conforming steps described above, the two schemas were brought to a Compatible state and became ready to proceed to the Merging & Restructuring stage.

4.2.4 Merging & Restructuring

At this stage, the Yasar University and Hacettepe University schemas, which were made compatible as a result of the Conforming processes, were overlaid and merged into a single logical schema using the superposition principle. As a result of this merging, the first Partial Integrated Schema (PIS) was obtained.

- Merging:

During the merging process, the following key decisions were applied:

Instructor Entities:
The FACULTY_MEMBER entity was adopted as the superclass in the global schema.
The instructor hierarchy (INSTRUCTOR, ASSISTANT, COORDINATOR) was preserved in accordance with the Yasar University model.

T+A+L / ECTS Attributes:
These attributes were removed from group-based entities (COMPULSORY, ELECTIVE_GROUP) and assigned as attributes to the COURSE entity in the global schema.

Concept Unlikeness Cases:
All missing attributes and relationships identified during the comparison stage were added to the relevant entities.
Within this scope:
- Missing attributes were added to the COURSE and ELECTIVE_GROUP entities,
- The LAB_FOR relationship was defined between COURSE – COURSE.

Following the merging of the schemas and the construction of the PIS schema, course concepts originating from different universities became capable of being related through explicit associations. At this stage, in order to clearly represent pairs of courses that are considered related based on the predefined comparison criteria, a MATCHES relationship was defined between COURSE and CURRICULUM_ITEM.

This approach allows the PIS schema to present a single, unified representation of course concepts while recording cross-university correspondences through relationship instances, without imposing a strict equivalence requirement at the schema level.

- Completeness & Correctness

As a result of the merging process, it was determined that all semantic information expressed by both schemas was preserved. During the conforming stage, the Yasar University instructor hierarchy (FACULTY_MEMBER, INSTRUCTOR, ASSISTANT, COORDINATOR), which provides a richer and more detailed structure, was preserved and integrated into the Hacettepe schema.

In addition, attributes and relationships that were missing in each schema were mutually added (for example, the LAB_FOR relationship from the Hacettepe schema and the Category and DeliveryMode attributes from the Yasar schema).

As a result of these adjustments:
- All information contained in both original schemas is represented in the PIS schema.
- No incorrect, incomplete, or conflicting information was introduced.
- In resolving conflicts, the representation that better preserves semantic integrity and provides greater completeness (e.g., associating the T+A+L / ECTS attributes with the COURSE entity) was taken as the basis.

- Minimality

It was observed that the same concept is not unnecessarily represented more than once in the global schema.

  - Structural Conflicts: The T+A+L / ECTS attributes were removed from the two group entities (COMPULSORY, ELECTIVE_GROUP) in the Hacettepe schema and assigned to a single main entity (COURSE).

Thus, unnecessary data redundancy was prevented. The standardization and superposition operations applied ensured that the global schema has a minimal structure.

- Understandability

After the merging and restructuring steps, it was evaluated that the resulting PIS schema gained a more understandable and sustainable structure.

- Standardization: In order to improve the understandability and semantic clarity of the global schema, the CURRICULUM_ITEM entity was restructured through a disjoint specialization into ELECTIVE_ITEM and COMPULSORY_ITEM. This restructuring eliminates the ambiguity caused by representing different curriculum item types within a single entity and ensures that relationships specific to elective and compulsory courses are clearly and consistently modeled.

- Hierarchy Clarity: The detailed instructor hierarchy adopted from the Yasar schema enabled clear and distinct representation of different roles (INSTRUCTOR, ASSISTANT, COORDINATOR).

- Sustainability: The applied syntactic restructuring steps increased the consistency of the schema, resulting in a structure that is more robust against future changes.

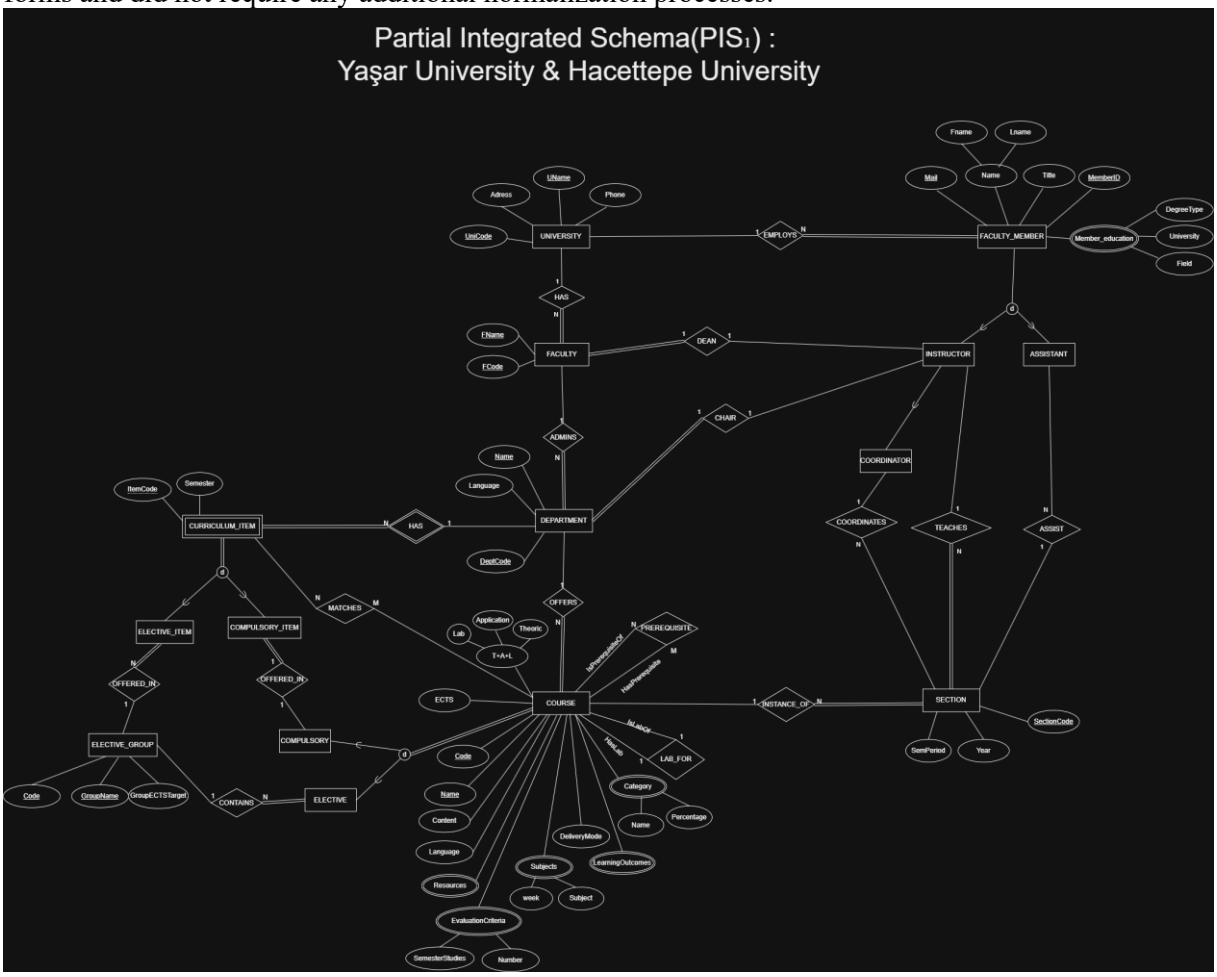
- Redundant Cycles Analysis

After merging, no redundant cycles that repeat information or introduce unnecessary complexity were identified.

- Normalization

After Integration As a result of the merging process, no significant normalization violations were observed that would disrupt the schema structure or require the definition of new entities.

Result:
Thanks to the conforming operations performed, the PIS schema preserved the basic normalization forms and did not require any additional normalization processes.



Overall Evaluation:
With the completion of this stage, the resulting PIS schema became ready for the second integration stage (PIS + Çukurova University EER).

Stage2: Partial Integrated Schema (PIS) - Cukurova University's EER

4.2.1 Comparison of Schemas

There are 20 differences between the two diagrams, which are detailed in the groups below. Apart from these, there are no structural or semantic differences in any of the other components.

4.2.1.1 Naming Conflicts

- Homonymous Naming Conflict Analysis

There isn't any homonymous naming conflict between two diagrams.

- Synonymous Naming Conflict Analysis

InstructorEducation attribute in Cukurova University's diagram has the same purpose as the MemberEducation attribute in the PIS.

InstructorId attribute in Cukurova University's diagram has the same purpose as the MemberId attribute in the PIS.

These are the synonymous naming conflicts between two diagrams.

4.2.1.2 Structural Conflicts

- Type Conflict Analysis:

No direct type conflict was observed; however, substantial restructuring was required due to semantic scope and hierarchy mismatches.

- Dependency Conflict Analyses:

  - In the Cukurova University schema, the ECTS and T+A attributes are stored in the COMPULSORY, DEPT_ELECTIVE, and OUTSIDE_ELECTIVE_GROUP entities, reflecting a design decision aimed at reducing data redundancy, as all outside elective courses share identical T+A and ECTS values. In contrast, the PIS schema assigns the ECTS and T+A+L attributes directly to the COURSE entity.

  - The T+A attribute in Cukurova University's schema includes only theoretical and application hours, whereas in the PIS schema it also covers laboratory hours (T+A+L). This difference is evaluated as a dependency conflict, since the same attribute represents different functional scopes across schemas. It also reflects a case of concept unlikeness, where the same-named concepts are semantically incompatible.

  - While the PIS's INSTRUCTOR entity is a subclass of the FACULTY_MEMBER entity, in Cukurova University's diagram INSTRUCTOR entity is the superclass. Because of that in the PIS the EMPLOYS relation is dependent on the FACULTY_MEMBER entity, but in the Cukurova University's diagram, it is dependent on the INSTRUCTOR entity.

  - The LAB_FOR relationship is associated with a subclass of the COURSE entity in the Çukurova University schema, whereas in the PIS schema it is defined as a relationship directly attached to the COURSE entity. This difference leads to a dependency conflict, since the relationship depends on different conceptual anchors across the schemas.

  - The CONTAINS relationship is associated with entities of different conceptual scope across the schemas. While it is linked to a more specialized entity in one schema, it is connected to a higher-level entity in the other. This situation constitutes a dependency conflict, as the relationship depends on different conceptual anchors.

  - In both schemas, the OFFERED_IN relationship is associated with different entities. This situation is evaluated as a dependency conflict, arising from the inconsistent definition of the conceptual dependency of the relationship.

- Key Conflict Analysis:

Comparing the primary key definitions revealed no key-level mismatches between the two schemes. Therefore, no further matching process was needed at this stage.

- Behavioral Conflict Analysis:

No differences were observed between the two schemas in terms of insert/delete behaviors. Accordingly, no behavioral conflict was identified.

- Concept Unlikeness:

    - In the PIS there is a FACULTY_MEMBER superclass and this superclass has INSTRUCTOR and ASSISTANT subclasses, but in Cukurova University's schema there isn't an ASSISTANT entity. So we don't have to make any specialization. In the Cukurova schema, all academic staff are modeled under the INSTRUCTOR concept. Because of that in Cukurova University's diagram, the INSTRUCTOR entity is the superclass.

    - In the Cukurova University's diagram, the OUTSIDE_ELECTIVE_GROUP is only for university electives. However, the ELECTIVE_GROUP in the PIS is broader because it includes both university and technical electives.

    - In Çukurova University's diagram, the ELECTIVE_GROUP concept is modeled using a specialization hierarchy consisting of the subclasses DEPT_ELECTIVE_GROUP and OUTSIDE_ELECTIVE_GROUP, whereas in PIS it is represented as a single flat ELECTIVE entity without any subclass structure. Although both representations refer to the same conceptual domain, they differ in their hierarchy modeling.

    - In Çukurova University's diagram, the ELECTIVE concept is modeled using a specialization hierarchy consisting of the subclasses DEPT_ELECTIVE and OUTSIDE_ELECTIVE, whereas in PIS it is represented as a single flat ELECTIVE entity without any subclass structure. Although both representations refer to the same conceptual domain, they differ in their dependency and hierarchy modeling.

    - In both schemas, the EVALUATION_CRITERIA are modeled as a multivalued attribute associated with the same entity. However, the semantic interpretation of this attribute differs across schemas, as the internal structure and the meanings of its components are not aligned. Additionally, the attribute's subcomponents are represented using different names, further contributing to semantic unlikeness at the attribute level.

    - In Cukurova University's diagram, the CURRICULUM_ITEM entity is divided into ELECTIVE_POOL_ITEM , ELECTIVE_SEMESTER_ITEM and COMPULSORY subclasses. On the other hand, the PIS CURRICULUM_ITEM entity is divided into ELECTIVE_ITEM and COMPULSORY_ITEM subclasses.

4.2.1.4 Other Modeling Differences Identified During Schema Comparison

- The GroupECTSTarget attribute, which is part of the ELECTIVE_GROUP entity in the PIS, is absent from the Cukurova University's diagram.

- The Category attribute in the PIS's COURSE entity has no equivalent attribute in Cukurova University's diagram.

- The DeliveryMode attribute in the PIS's COURSE entity has no equivalent attribute in Cukurova University's diagram.

- During the schema comparison phase, it was observed that the integrated schema obtained in the previous step includes a MATCHES relationship to represent course correspondences, whereas the Çukurova University schema does not define any explicit relationship for modeling course-level equivalencies.

- This difference does not constitute a naming or structural conflict, since the absence of such a relationship in the Çukurova schema does not introduce a semantic contradiction. Rather, it reflects a modeling scope difference, where course correspondences are not explicitly represented in the original schema.

- The recursive PREREQUISITE relationship is explicitly modeled in the PIS schema to represent prerequisite constraints between courses. However, this relationship does not exist in Çukurova University's schema. This difference indicates a structural unlikeness arising from the absence of a relationship in one of the schemas, rather than a semantic or naming conflict.

4.2.3 Conforming

At this stage, the structural and semantic conflicts identified during the comparison process were resolved using the conforming methods proposed in the literature.The applied solutions are presented below under subheadings.

- Naming Conflict Resolutions:
    - The name of the InstructorEducation attribute in Cukurova University's diagram is changed to MemberEducation.
    - The name of the InstructorId attribute in Cukurova University's diagram is changed to MemberId.

- Concept Unlikeness Resolutions
    - Instructor Subclass Hierarchy Solution:
      In terms of the hierarchical structure related to the instructor concept, it was determined that the PIS provides a richer and more detailed model. In accordance with the completeness principle, this structure was preserved for the global schema. For this reason, syntactic restructuring was applied to the Cukurova University schema, and the following modifications were made:

      The FACULTY_MEMBER superclass was added to the Cukurova schema.
    - The existing INSTRUCTOR entity was moved under this superclass.
      The ASSISTANT subclass was added in PIS as a disjoint specialization alongside INSTRUCTOR under the FACULTY_MEMBER entity.
      The ASSISTANT entity was connected to the SECTION entity via the ASSIST relationship.
      Roles derived from INSTRUCTOR in the Cukurova schema (e.g., DEAN, CHAIR) were preserved as INSTRUCTOR subclasses in accordance with the PIS.

      With these modifications, the instructor concept in the two schemas was made hierarchically consistent and integrated.

    - To resolve the semantic unlikeness in the EVALUATION_CRITERIA attribute, the structure defined in Çukurova University's schema was preserved as the reference

model. During the conforming phase, the representation in PIS was modified to align with the internal structure and semantic definitions used in the Çukurova schema.

- To resolve the hierarchy-level unlikeness observed in the representation of ELECTIVE_GROUP and ELECTIVE concepts, the modeling approach used in Çukurova University's schema was preserved as the reference structure. During the conforming phase, the flat representations in PIS were restructured to align with the specialization hierarchies defined in the Çukurova schema. Accordingly, the ELECTIVE_GROUP concept in PIS was reorganized into the subclasses DEPT_ELECTIVE_GROUP and OUTSIDE_ELECTIVE_GROUP, and the ELECTIVE concept was similarly refined into DEPT_ELECTIVE and OUTSIDE_ELECTIVE. This conforming operation ensures consistent hierarchy modeling across schemas while preserving the original semantic distinctions of elective-related concepts.

- During the conforming phase, the specialization hierarchy in PIS was refined to align with the more granular structure defined in the Çukurova schema. Accordingly, ELECTIVE_ITEM in PIS was restructured into two subclasses, ELECTIVE_POOL_ITEM and ELECTIVE_SEMESTER_ITEM, while preserving COMPULSORY_ITEM as the compulsory subtype.

- Dependency Conflict Resolutions:
  - During the conforming phase, this issue was resolved by redefining the CONTAINS relationship to operate between the newly introduced elective-related subclasses. Accordingly, CONTAINS was consistently associated with DEPT_ELECTIVE_GROUP – DEPT_ELECTIVE and OUTSIDE_ELECTIVE_GROUP – OUTSIDE_ELECTIVE, aligning the dependency structure with the refined specialization hierarchies. This conforming operation eliminates ambiguity in relationship dependencies while preserving the original semantics of elective containment.

  - During the conforming stage, the dependency conflict caused by linking the OFFERED_IN relationship to different entities was resolved by unifying its conceptual dependency. OFFERED_IN was redefined through the appropriate subclasses of CURRICULUM_ITEM; endpoints previously attached to different entities were detached and reassigned to the correct abstraction level. As a result, OFFERED_IN was standardized as COMPULSORY_ITEM–COMPULSORY, ELECTIVE_POOL_ITEM–OUTSIDE_ELECTIVE_GROUP, and ELECTIVE_SEMESTER_ITEM–DEPT_ELECTIVE_GROUP, ensuring consistent dependency structures across the schemas.

  - Solution of EMPLOYS relation's difference:
    Since all faculty members are employed by the institution, the EMPLOYS relation is logically a property that applies to all faculty members. Therefore, applying the generalization principle to Cukurova University's diagram, the dependency of the EMPLOYS relation is detached from the INSTRUCTOR entity, which was previously a superclass and then subclassed by FACULTY_MEMBER, and attached to the superclass FACULTY_MEMBER entity.

  - T+A+L / ECTS Dependency Conflict Resolution:
    Considering the dependency conflict between the T+A and T+A+L attributes, the T+A+L structure was selected to be preserved. In this context, T+A is a composite attribute consisting of theoretical and application hours, whereas T+A+L is a composite attribute that additionally includes laboratory hours. Accordingly, the T+A

attributes in the Cukurova University schema were transformed into the more comprehensive composite T+A+L attribute, ensuring a consistent dependency structure for workload representation. To maintain consistency in attribute dependencies, the T+A+L attribute was retained only at the superclass level, and redundant T+A+L definitions in the subclasses were removed.

- As a result, the T+A+L structure in the Cukurova University schema was relocated to the COURSE entity, aligning the workload representation with the appropriate conceptual level.

- LAB_FOR dependency conflict Resolution: In the PIS, the LAB_FOR relationship is linked to the more encompassing COURSE entity, and this structure is used to maintain semantic consistency in both schemas. In Cukurova University, The LAB_FOR relationship, which is linked to the COMPULSORY subclass, is assigned to the more encompassing COURSE entity.

- Addition of schema extension Attributes and Relationships

  - Missing GroupECTSTarget resolution: A GroupECTSTarget attribute should be added to the new ELECTIVE_GROUP entity in Cukurova University's diagram to preserve the meaning of both schemas .

  - Missing Lab attribute in T+A resolution: To avoid data loss, the richer T+A+L attribute found in the PIS is used instead of the T+A attribute at Cukurova University.

  - Missing Category attribute resolution: A Category attribute should be added to the Cukurova University diagram's COURSE entity to avoid loss of meaning.

  - Missing DeliveryMode attribute resolution: A DeliveryMode attribute should be added to the COURSE entity in Cukurova University's diagram to preserve the meaning of both schemas.

  - The recursive PREREQUISITE relationship, which was absent in Çukurova University's schema, was introduced in the Cukurova University's diagram by associating it with the COURSE entity.

  - During the conforming phase, the absence of an explicit relationship for representing course correspondences in the Çukurova University schema was addressed through schema extension. Since this difference does not introduce a semantic contradiction, it was treated as a modeling scope variation rather than a structural conflict. Accordingly, the integrated model was prepared to include a MATCHES association to uniformly represent cross-university course correspondences.

4.2.4 Merging & Restructuring

At this stage, the Cukurova University schema and the Partial Integrated Schema (PIS), which were made compatible as a result of the Conforming processes, were overlaid and merged into a single logical schema using the superposition principle. As a result of this merging process, the Global Integrated Schema was obtained.

- Merging

In this step, the conformed schemas were combined by unifying entities, relationships, and attributes that represent equivalent concepts.

The specialization hierarchies and dependency structures defined during the conforming stage were directly incorporated into the merged schema without introducing new conflicts.
The resulting schema formed the basis for the subsequent restructuring and evaluation steps, which assess the quality of the integration according to established criteria.

- Completeness & Correctness

Thanks to the conforming operations performed prior to merging, all entities, relationships, and attributes from both schemas were successfully preserved.Missing attributes such as GroupECTSTarget, Category, and DeliveryMode were explicitly added, and richer attribute structures (e.g., TAL) were retained.No information loss or incorrect abstraction was observed, indicating that the resulting global schema satisfies the completeness and correctness criterion.

- Minimality

During the merging process, redundant representations caused by parallel modeling choices were eliminated.As a result, each concept is represented exactly once in the global schema, satisfying the minimality requirement.

- Understandability: The resulting global schema is easy to understand, since concepts are represented with standardized names and clear hierarchical structures.

- Standardization: Although the merged structure correctly preserved the underlying semantics, the original names derived from institution-specific modeling were found to be less expressive at the global level. Therefore, the following renaming operations were performed to provide a more meaningful and intuitive representation:
    - OUTSIDE_ELECTIVE_GROUP was renamed to
    - POOL_ELECTIVE_GROUP
    - OUTSIDE_ELECTIVE was renamed to POOL_ELECTIVE
    - DEPT_ELECTIVE_GROUP was renamed to SEMESTER_ELECTIVE_GROUP
    - DEPT_ELECTIVE was renamed to SEMESTER_ELECTIVE

  This renaming does not alter the structural or semantic integrity of the schema but enhances understandability by reflecting the actual academic organization of elective courses more accurately.

- Hierarchy Clarity: The hierarchical structures within the schema were reorganized to clearly reflect conceptual relationships. The FACULTY_MEMBER superclass and its specializations) provide a clear and intuitive representation of academic staff roles.

- Sustainability:The applied restructuring decisions improved the long-term maintainability of the schema.

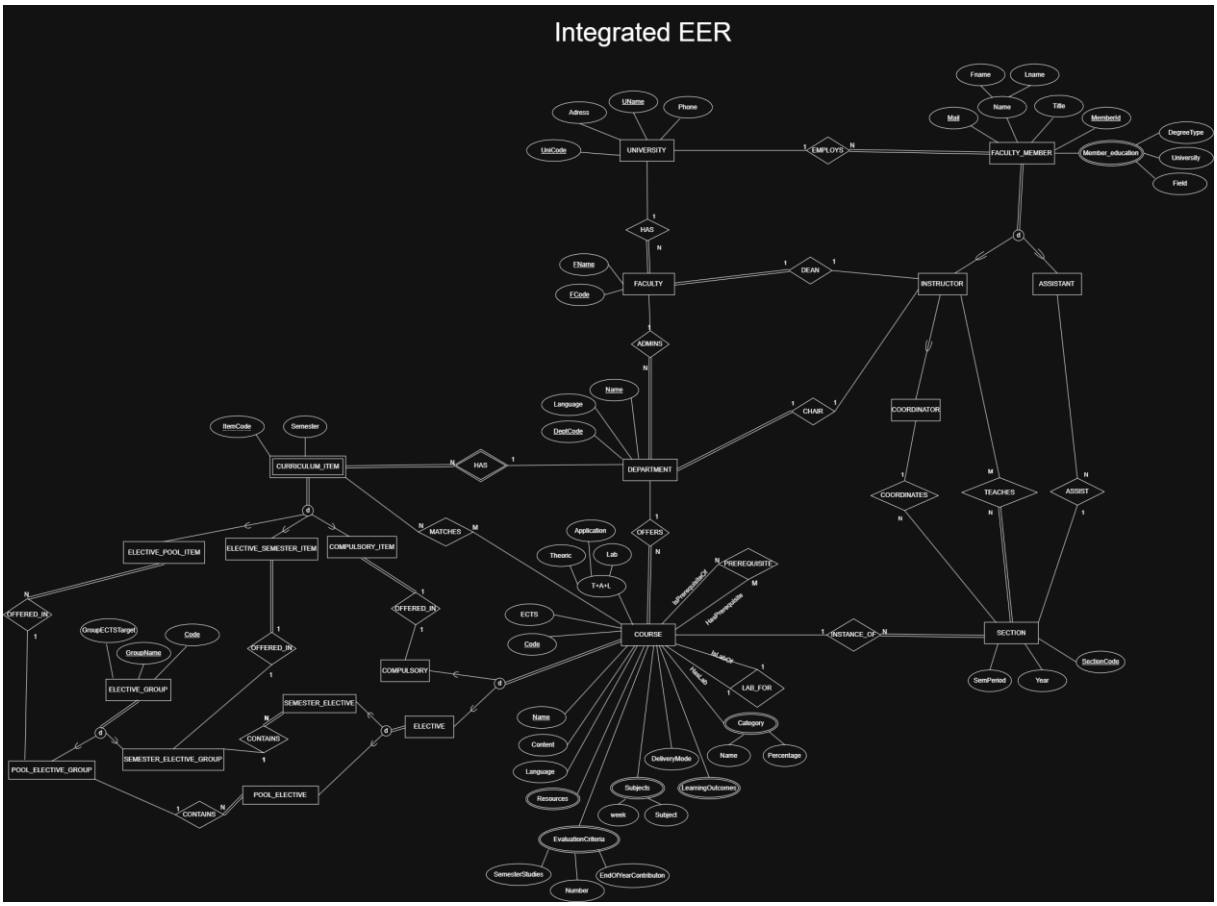
- Redundant Cycles Analysis

After merging, no redundant cycles that repeat information or introduce unnecessary complexity were identified.

- Normalization

After Integration No obvious normalization issues were observed in the integrated EER; potential cross-schema functional dependencies were checked and no restructuring was deemed necessary.

Result: Thanks to the conforming operations performed, the Global Integrated Schema preserved the basic normalization forms and did not require any additional normalization processes.



The integration process was finalized with the formation of a global integrated schema.

5. Logical Model

5.1 Mapping

ITERATION 1

Step 1:

UNIVERSITY(<u>UniCode</u>, Address, Name, Phone)

FACULTY (<u>Fcode</u>, Fname)

DEPARTMENT(<u>DeptCode</u>, Name, Language)

SECTION(<u>SectionCode</u>, SemPeriod, Year)

Step 2:

CURRICULUM_ITEM(<u>DEPARTMENT.DeptCode, ItemCode</u>,Semester) // via HAS

Step 3 :There isn't any 1:1 relation in this iteration.

Step 4:

FACULTY(..., UNIVERSITY.UniCode) // via HAS

DEPARTMENT(..., FACULTY.Fcode) // via ADMINS

Step 5 : There isn't any M:N relationship for this iteration.

Step 6: There isn't any multivalued attribute for this iteration.

Step 7: There isn't any N-ary relationship.

Step 8:

8c) COURSE(Code, Name, ECTS, T+A+L.Theoric, T+A+L.Application, T+A+L.Labratory, Content, Language, DeliveryMode,CourseType)

8c) FACULTY_MEMBER(MemberId, Title, Name.Fname, Name.Lname, Mail, MemberType)

8a)
ELECTIVE_POOL_ITEM(DEPARTMETN.DeptCode, ItemCode) // via CURRICULUM_ITEM

ELECTIVE_SEMESTER _ITEM(DEPARTMETN.DeptCode.CurriculumCode, ItemCode) // via CURRICULUM_ITEM

COMPULSORY_ITEM(DEPARTMETN.DeptCode, ItemCode) // via CURRICULUM_ITEM

8c) ELECTIVE_GROUP(GroupCode, GroupName, GroupECTSTarget,GroupType)

Step 9: There isn't any union type.

ITERATION 2

Step 1: There isn't any regular entity.

Step 2: There isn't any weak entity.

Step 3:

FACULTY(..., FACULTY_MEMBER.MemberId) // via DEAN

DEPARTMENT(..., FACULTY_MEMBER.MemberId) // via CHAIR

COURSE(..., COURSE.LabId) // via LAB_FOR

COMPULSORY_ITEM(...,COURSE.CourseCode) // via OFFERED_IN

ELECTIVE_SEMESTER_ITEM(..., ELECTIVE_GROUP.GroupCode)// via OFFERED_IN

Step 4:

FACULTY_MEMBER(..., UNIVERSITY.UniCode) // via EMPLOYS

SECTION(..., COURSE.CourseCode) // via INSTANCE_OF

SECTION(..., FACULTY_MEMBER.MemberId) // via ASSIST

COURSE(..., DEPARTMENT.DeptCode) // via OFFERS

ELECTIVE_POOL_ITEM(...,ELECTIVE_GROUP.Code) // via OFFERED_IN

Step 5:

PREREQUISITE(COURSE.CourseCode, COURSE.PrerequisiteCode) // via PREREQUISITE

TEACHES(SECTION.SectionCode,FACULTY_MEMBER.InstructorId) // via TEACHES

MATCHES(DEPARTMENT.DeptCode,CURRICULUM_ITEM.ItemCode,COURSE.Code)// via MATCHES

Step 6:

MEMBER_EDUCATION(FACULTY_MEMBER.MemberId, DegreeType, University, Field) // via FACULTY_MEMBER

RESOURCES(COURSE.CourseCode, Resource) // via COURSE

EVALUATION_CRITERIA(COURSE.CourseCode, SemesterStudies, Number, ContributionDuringYear, EndOfYearContributon) // via COURSE

SUBJECTS(COURSE.CourseCode, Week, Subject) // via COURSE

LEARNING_OUTCOMES(COURSE.CourseCode, Outcome) // via COURSE

CATEGORY(COURSE.CourseCode, Name, Percentage) // via COURSE

Step 7: There isn't any N-ary relationship.

Step 8:

8c) COURSE(...,ElectiveType)

8c) FACULTY_MEMBER(..., IsCoordinator)

Step 9: There isn't any union type.

ITERATION 3

Step 1: There isn't any regular entity.

Step 2: There isn't any weak entity.

Step 3: There isn't any 1:1 relationship.

Step 4:

COURSE(..., ELECTIVE_GROUP.Code) // via CONTAINS

SECTION(..., FACULTY_MEMBER.MemberId) // via COORDINATES

Mapping is completed.

5.2 Relational Schema

UNIVERSITY

UniCode	Address	Name	Phone

FACULTY_MEMBER

MemberId	Title	Fname	Lname
UniCode	MemberType	IsCoordinator	Mail

FACULTY

Fcode	Fname	UniCode	DeanId

DEPARTMENT

DeptCode	Name	Language	FacultyCode	ChairId

MEMBER_EDUCATION

MemberId	DegreeType	University	Field

ELECTIVE_GROUP

GroupCode	GroupName	GroupECTSTarget	GroupType

COURSE

Code	Name	ECTS	Theoric	Application
Laboratory	Language	DeliveryMode	LabCode	DeptCode
CourseType	Content	ElectiveType	ElectiveGroup	

SECTION

SectionCode	SemPeriod	Year	CourseCode	CoordinatorId	AsistantId

TEACHES

SectionCode	InstructorId

CURRICULUM_ITEM

ItemCode	DeptCode	Semester	CurriculumItemType	GroupOrCourseCode

ELECTIVE_SEMESTER_ITEM

DeptCode	ItemCode	ElectiveGroupCode

ELECTIVE_POOL_ITEM

DeptCode	ItemCode	ElectiveGroupCode

COMPULSORY_ITEM

DeptCode	ItemCode	CompulsoryCode

MATCHES

DeptCode	ItemCode	CourseCode

RESOURCES

ResourceId	CourseCode	Resource

CATEGORY

CategoryId	CourseCode	Name	Percentage

PREREQUISITE

CourseCode	PrerequisiteCode

LEARNING_OUTCOMES

OutcomeId	CourseCode	Outcome

SUBJECTS

CourseCode	Week	Subject

EVALUATION_CRITERIA

EvalCriteriaId	CourseCode	SemesterStudies	Number	Contributon

6. Physical Implementation and Querying

6.1 DBMS Selection

PostgreSQL was used as the database management system for implementing and validating the integrated schema in this project.

6.2 Database Schema Definition (SQL DDL Scripts)

```
CREATE TABLE UNIVERSITY(
	UniCode VARCHAR(4) PRIMARY KEY,
	Address VARCHAR(100) NOT NULL,
	Name VARCHAR(30) NOT NULL UNIQUE,
	Phone VARCHAR(13) UNIQUE,

	CHECK (Phone IS NULL OR Phone ~ '^(0[0-9]{10}|\+90[0-9]{10})$')
);

CREATE TABLE FACULTY_MEMBER(
	MemberId INT PRIMARY KEY,
	Title VARCHAR(20),
	Fname VARCHAR(20) NOT NULL,
	Lname VARCHAR(20) NOT NULL,
	Mail VARCHAR(30) UNIQUE,
	UniCode VARCHAR(4) NOT NULL
		REFERENCES UNIVERSITY(UniCode)
		ON DELETE RESTRICT
		ON UPDATE RESTRICT,
	MemberType CHAR(2) NOT NULL,
	IsCoordinator BOOLEAN NOT NULL,

	CONSTRAINT chk_member_type
	CHECK (MemberType IN ('IN', 'AS')),

	-- if member is Assistant, then Coordinator FALSE
	CONSTRAINT chk_assistant_coordinator
	CHECK (NOT (MemberType = 'AS' AND IsCoordinator = TRUE))
);

CREATE TABLE FACULTY (
	Fcode VARCHAR(8) PRIMARY KEY,
	Fname VARCHAR(30) NOT NULL,
	UniCode VARCHAR(4) NOT NULL
		REFERENCES UNIVERSITY(UniCode)
		ON DELETE RESTRICT
		ON UPDATE CASCADE,
	DeanId INT NOT NULL UNIQUE DEFAULT 0
		REFERENCES FACULTY_MEMBER(MemberId)
		ON DELETE SET DEFAULT
		ON UPDATE CASCADE
);
```

```
CREATE TABLE DEPARTMENT(
	DeptCode VARCHAR(15) PRIMARY KEY,
	Name VARCHAR(40) NOT NULL,
	Language VARCHAR(15) NOT NULL,
	FacultyCode VARCHAR(8) NOT NULL
		REFERENCES FACULTY(Fcode)
	ON DELETE RESTRICT
		ON UPDATE CASCADE,
	ChairId INT NOT NULL UNIQUE DEFAULT 0
		REFERENCES FACULTY_MEMBER(MemberId)
		ON DELETE SET DEFAULT
		ON UPDATE CASCADE
);

CREATE TABLE MEMBER_EDUCATION(
	MemberId INT
		REFERENCES FACULTY_MEMBER(MemberId)
		ON DELETE CASCADE
		ON UPDATE CASCADE,
	DegreeType VARCHAR(50),
	University VARCHAR(100),
	Field VARCHAR(150),

	CONSTRAINT pk_member_education
	PRIMARY KEY (MemberId, DegreeType, University, Field)
);

CREATE TABLE ELECTIVE_GROUP(
	GroupCode VARCHAR(20) PRIMARY KEY,
	GroupName VARCHAR(50) NOT NULL,
	GroupECTSTarget INT,
	GroupType VARCHAR(13) NOT NULL,

	CONSTRAINT check_elective_group_type
		CHECK (GroupType IN ('SEM_ELECT_GR', 'POOL_ELECT_GR'))
);

CREATE TABLE COURSE(
	Code VARCHAR(10) PRIMARY KEY,
	Name VARCHAR(150) NOT NULL,
	ECTS INT NOT NULL,
	Theoric INT,
	Application INT,
	Laboratory INT,
	Language VARCHAR(15),
	DeliveryMode VARCHAR(8) DEFAULT 'On-Site',
	LabCode VARCHAR(10)
		REFERENCES COURSE(Code),
	DeptCode VARCHAR(15)
		REFERENCES DEPARTMENT(DeptCode)
	ON DELETE RESTRICT
```

```
		ON UPDATE CASCADE,
	CourseType VARCHAR(11) NOT NULL,
	Content TEXT,
	ElectiveType VARCHAR(20),
	ElectiveGroup VARCHAR(30)
		REFERENCES ELECTIVE_GROUP(GroupCode)
		ON DELETE CASCADE
		ON UPDATE CASCADE,

	CONSTRAINT chk_elective_type
	CHECK (ElectiveType IN ('Pool Elective', 'Semester Elective')),
	CONSTRAINT chk_course_type
	CHECK (CourseType IN ('Elective', 'Compulsory'))

);

-- It could have been inside CREATE TABLE, but we wrote it here to use ALTER TABLE.
ALTER TABLE COURSE
ADD CONSTRAINT chk_course_total_hours
CHECK (
	COALESCE(Theoric, 0)
  + COALESCE(Application, 0)
  + COALESCE(Laboratory, 0) > 0
);

-- If ElectiveType = 'Pool Elective', then ElectiveGroup can be any group in the ELECTIVE_GROUP table
-- If ElectiveType = 'Semester Elective', then ElectiveGroup must be NULL
-- If it's Compulsory, then ElectiveType = NULL and ElectiveGroup = NULL
ALTER TABLE COURSE
ADD CONSTRAINT chk_course_elective_logic
CHECK (
	(
	CourseType = 'Compulsory'
	AND ElectiveType IS NULL
	AND ElectiveGroup IS NULL
	)
	OR
	(
	CourseType = 'Elective'
	AND ElectiveType = 'Pool Elective'
	AND ElectiveGroup IS NOT NULL
	)
	OR
	(
	CourseType = 'Elective'
	AND ElectiveType = 'Semester Elective'
	AND ElectiveGroup IS NOT NULL
	)
);
```

```
CREATE TABLE SECTION(
	SectionCode VARCHAR(15) PRIMARY KEY,
	SemPeriod VARCHAR(6) NOT NULL,
	Year SMALLINT NOT NULL,
	CourseCode VARCHAR(10)
		REFERENCES COURSE(Code)
		ON DELETE CASCADE
		ON UPDATE CASCADE,
	CoordinatorId INT
		REFERENCES FACULTY_MEMBER(MemberId)
		ON DELETE SET NULL
		ON UPDATE CASCADE,
	AssistantId INT
		REFERENCES FACULTY_MEMBER(MemberId)
		ON DELETE SET NULL
		ON UPDATE CASCADE,

	CONSTRAINT chk_sem_period
	CHECK (SemPeriod IN ('FALL', 'SPRING')),

	-- The same course cannot be offered more than once in the same semester.
	CONSTRAINT uniq_section_course_term
		UNIQUE (CourseCode, Year, SemPeriod)
);

CREATE TABLE TEACHES(
	SectionCode VARCHAR(15)
		REFERENCES SECTION(SectionCode)
		ON DELETE CASCADE
		ON UPDATE CASCADE,
	InstructorId INT
		REFERENCES FACULTY_MEMBER(MemberId)
		ON DELETE CASCADE
		ON UPDATE CASCADE,

	PRIMARY KEY(SectionCode, InstructorId)
);

CREATE TABLE CURRICULUM_ITEM(
	ItemCode SERIAL,
	DeptCode VARCHAR(15)
		REFERENCES DEPARTMENT(DeptCode)
		ON DELETE RESTRICT
		ON UPDATE RESTRICT,
	Semester SMALLINT NOT NULL,
	CurriculumItemType VARCHAR(7) NOT NULL,
	GroupOrCourseCode VARCHAR(15) NOT NULL,
	PRIMARY KEY (DeptCode, ItemCode),
	CONSTRAINT chk_semester
	CHECK (Semester IN (1,2,3,4,5,6,7,8)),

	CONSTRAINT chk_curriculum_item_type
```

```
                CHECK (CurriculumItemType IN ('EL-POOL', 'EL-SEM', 'COMPUL'))
);

CREATE TABLE ELECTIVE_POOL_ITEM(
        DeptCode VARCHAR(15),
        ItemCode INT,
        ElectiveCode VARCHAR(15)
                REFERENCES ELECTIVE_GROUP(GroupCode)
                ON DELETE RESTRICT
                ON UPDATE CASCADE,

        PRIMARY KEY (DeptCode, ItemCode),
        FOREIGN KEY (DeptCode, ItemCode)
        REFERENCES CURRICULUM_ITEM(DeptCode, ItemCode)
        ON DELETE RESTRICT
                ON UPDATE CASCADE
);

CREATE TABLE ELECTIVE_SEMESTER_ITEM(
        DeptCode VARCHAR(15),
        ItemCode INT,
        ElectiveCode VARCHAR(15) UNIQUE
                REFERENCES ELECTIVE_GROUP(GroupCode)
                ON DELETE RESTRICT
                ON UPDATE CASCADE,

        PRIMARY KEY (DeptCode, ItemCode),
        FOREIGN KEY (DeptCode, ItemCode)
        REFERENCES CURRICULUM_ITEM(DeptCode, ItemCode)
        ON DELETE RESTRICT
                ON UPDATE CASCADE
);

CREATE TABLE COMPULSORY_ITEM(
        DeptCode VARCHAR(15),
        ItemCode INT,
        CompulsoryCode VARCHAR(15) UNIQUE
        REFERENCES COURSE(Code)
        ON DELETE RESTRICT
        ON UPDATE CASCADE,

        PRIMARY KEY (DeptCode, ItemCode),
        FOREIGN KEY (DeptCode, ItemCode)
        REFERENCES CURRICULUM_ITEM(DeptCode, ItemCode)
        ON DELETE RESTRICT
                ON UPDATE CASCADE
);

CREATE TABLE MATCHES(
        DeptCode VARCHAR(15),
        ItemCode INT,
        CourseCode VARCHAR(10),
```

```
	PRIMARY KEY(DeptCode, ItemCode, CourseCode),
	FOREIGN KEY(DeptCode,ItemCode)
	REFERENCES CURRICULUM_ITEM(DeptCode,ItemCode)
	ON DELETE CASCADE
	ON UPDATE CASCADE,

	FOREIGN KEY(CourseCode)
	REFERENCES COURSE(Code)
	ON DELETE CASCADE
	ON UPDATE CASCADE
);

CREATE TABLE RESOURCES(
	ResourceId SERIAL PRIMARY KEY,
	CourseCode VARCHAR(10)
		REFERENCES COURSE(Code)
		ON DELETE CASCADE
		ON UPDATE CASCADE,
	Resource VARCHAR(300)
);

CREATE TABLE CATEGORY(
	CategoryId SERIAL PRIMARY KEY,
	CourseCode VARCHAR(10)
		REFERENCES COURSE(Code)
		ON DELETE CASCADE
		ON UPDATE CASCADE,
	Name VARCHAR(40),
	Percentage INT
);

CREATE TABLE PREREQUISITE(
	CourseCode VARCHAR(10)
		REFERENCES COURSE(Code)
		ON DELETE CASCADE
		ON UPDATE CASCADE,
	PrerequisiteCode VARCHAR(10)
		REFERENCES COURSE(Code)
		ON DELETE CASCADE
		ON UPDATE CASCADE,

	CONSTRAINT pk_prerequisite
	PRIMARY KEY (CourseCode, PrerequisiteCode)
);

CREATE TABLE LEARNING_OUTCOMES(
	OutcomeId SERIAL PRIMARY KEY,
	CourseCode VARCHAR(10)
		REFERENCES COURSE(Code)
		ON DELETE CASCADE
		ON UPDATE CASCADE,
```

```
	Outcome TEXT
);

CREATE TABLE SUBJECTS(
	CourseCode VARCHAR(10)
		REFERENCES COURSE(Code)
		ON DELETE CASCADE
		ON UPDATE CASCADE,
	Week INT,
	Subjects TEXT,
	CONSTRAINT pk_subjects
	PRIMARY KEY (CourseCode, Week)
);

CREATE TABLE EVALUATION_CRITERIA(
	EvalCriteriaId SERIAL PRIMARY KEY,
	CourseCode VARCHAR(10)
		REFERENCES COURSE(Code)
		ON DELETE CASCADE
		ON UPDATE CASCADE,
	SemesterStudies VARCHAR(50),
	Number INT DEFAULT 1,
	Contribution INT
);
```

6.3 Sample Data Population (SQL Scripts)

```
INSERT INTO UNIVERSITY (UniCode, Address, Name, Phone)
VALUES
	('YAUN', 'Selçuk Yaşar Kampüsü, Kazımdirik, Üniversite Cd. Ağaçlı Yol No: 37-39, 35100
Bornova/İzmir', 'YASAR UNIVERSITY', '02325707070'),
	('HAUN', 'Hacettepe Üniversitesi Sıhhiye, 06100, Ankara', 'HACETTEPE UNIVERSITY',
'03123052141'),
	('CUUN', 'Balcalı Kampüsü, 01330 Sarıçam/ADANA', 'CUKUROVA UNIVERSITY',
'+903223386623'),
	('UNKN', 'Unknown', 'Unknown', NULL);

INSERT INTO FACULTY_MEMBER
(MemberId,Title,Fname,Lname,Mail,Unicode,MemberType,IsCoordinator)
VALUES
	(0, 'Unknown', 'Unknown', 'Unknown', ' ', 'UNKN', 'IN', FALSE),
	(1, 'Assistant Professor', 'Dindar', 'Öz', 'dindar.oz@yasar.edu.tr', 'YAUN', 'IN', TRUE),
	(2, 'Research Assist', 'Tolga Buğra', 'Altuntaş', NULL, 'YAUN', 'AS', FALSE),
	(3, 'Associate Professor', 'Mete', 'Eminağaoğlu', 'mete.eminagaoglu@yasar.edu.tr', 'YAUN', 'IN',
FALSE),
	(4, 'Assistant Professor', 'Umut', 'Avcı', 'umut.avcı@yasar.edu.tr', 'YAUN', 'IN', TRUE),
	(5, 'Research Assist', 'Atabarış', 'Ayaydın', 'atabaris.ayaydin@yasar.edu.tr', 'YAUN', 'AS', FALSE),
	(6, 'Assistant Professor', 'Cansu Aksu', 'Raffard', 'cansu.aksu@yasar.edu.tr', 'YAUN', 'IN', TRUE),
	(7, 'Lecturer', 'Nejla', 'Gündoğar', 'nejla.gundogar@yasar.edu.tr', 'YAUN', 'IN', FALSE),
	(8, 'Associate Professor', 'Burhan', 'Gülbahar', 'burhan.gulbahar@yasar.edu.tr', 'YAUN', 'IN', TRUE),
	(9, 'Associate Professor', 'Nalan', 'Özkurt', 'nalan.ozkurt@yasar.edu.tr', 'YAUN', 'IN', FALSE),
```

```
(10, 'Assistant Professor', 'Fuat', 'Akal', 'akal@hacettepe.edu.tr', 'HAUN', 'IN', FALSE),
(11, 'Professor', 'Mehmet Erkut', 'Erdem', 'erkut@hacettepe.edu.tr', 'HAUN', 'IN', FALSE),
(12, 'Professor', 'Suat', 'Özdemir', 'suatozdemir@hacettepe.edu.tr', 'HAUN', 'IN', FALSE),
(13, 'Associate Professor', 'Harun', 'Artuner', 'artuner@hacettepe.edu.tr', 'HAUN', 'IN', FALSE),
(14, 'Professor', 'Zekeriya', 'Tüfekçi', 'ztufekci@cu.edu.tr', 'CUUN', 'IN', TRUE),
(15, 'Associate Professor', 'Fatih', 'Abut', 'fabut@cu.edu.tr', 'CUUN', 'IN', TRUE),
(16, 'Assistant Professor', 'Elif Emel', 'Fırat', 'eefirat@cu.edu.tr', 'CUUN', 'IN', TRUE),
(17, 'Lecturer', 'Yunus Emre', 'Çoğurcu', 'ycogurcu@cu.edu.tr', 'CUUN', 'IN', FALSE),
(18, 'Lecturer', 'Murat', 'Kara', 'mkara@cu.edu.tr', 'CUUN', 'IN', TRUE),
(19, 'Professor', 'Ayhan Özgür', 'Toy', 'ozgur.toy@yasar.edu.tr', 'YAUN', 'IN', FALSE),
(20, 'Associate Professor', 'Korhan', 'Karabulut', 'korhan.karabulut@yasar.edu.tr', 'YAUN', 'IN',
FALSE),
(21, 'Professor', 'Halil', 'Vural', 'ghalil@hacettepe.edu.tr', 'HAUN', 'IN', FALSE),
(22, 'Professor', 'Ali', 'Keskin', 'akeskin@cu.edu.tr', 'CUUN', 'IN', FALSE),
(24, 'Professor', 'Umut', 'Orhan', 'uorhan@cu.edu.tr', 'CUUN', 'IN', TRUE),
(25, 'Professor', 'Sevil', 'Şen', 'sevilsen@hacettepe.edu.tr', 'HAUN', 'IN', FALSE),
(26, 'Associate Professor', 'Cemil', 'zalluhoğlu', 'cemil@cs.hacettepe.edu.tr' , 'HAUN', 'IN', FALSE),
(27, 'Professor', 'Tunca', 'Doğan', 'tuncadogan@hacettepe.edu.tr', 'HAUN', 'IN', FALSE),
(28, 'Professor', 'İlyas', 'Çilekli', NULL, 'HAUN', 'IN', FALSE),
(29, 'Professor', 'Pınar Duygulu', 'Şahin', 'pinar.duygulu@hacettepe.edu.tr', 'HAUN', 'IN', FALSE),
(30, 'Professor', 'Ayça', 'Kolukısa', 'ayca.tarhan@hacettepe.edu.tr', 'HAUN', 'IN', FALSE),
(31, 'Assistant Professor', 'Arman', 'Savran', 'arman.savran@yasar.edu.tr', 'YAUN', 'IN', TRUE),
(32, 'Assistant Professor', 'Kazım', 'Erdoğdu', 'kazim.erdogdu@yasar.edu.tr', 'YAUN', 'IN', FALSE),
(33, 'Assistant Professor', 'Barış', 'Yıldız', 'baris.yildiz@yasar.edu.tr', 'YAUN', 'IN', TRUE),
(34, 'Assistant Professor', 'Deniz', 'Özsoyeller', 'deniz.ozsoyeller@yasar.edu.tr', 'YAUN', 'IN', TRUE),
(35, 'Assistant Professor', 'Uras', 'Tos', 'uras.tos@yasar.edu.tr', 'YAUN', 'IN', FALSE),
(36, 'Lecturer', 'Asadul', 'Islam', 'asadul.islam@yasar.edu.tr', 'YAUN', 'IN', TRUE),
(37, 'Lecturer', 'Filiz Pars', 'Uçağı', 'filiz.ucagi@yasar.edu.tr', 'YAUN', 'IN', TRUE),
(38, 'Lecturer', 'Elvira', 'Korukcu', 'elvira.korukcu@yasar.edu.tr', 'YAUN', 'IN', FALSE),
(39, 'Lecturer', 'Cenk', 'Erdem', 'cenk.erdem@yasar.edu.tr', 'YAUN', 'IN', FALSE),
(40, 'Professor', 'Ahmet Hasan', 'Koltuksuz', 'ahmet.koltuksuz@yasar.edu.tr', 'YAUN', 'IN', TRUE),
(41, 'Assistant Professor', 'Barış', 'Ata', 'bata@cu.edu.tr', 'CUUN', 'IN', TRUE),
(42, 'Professor', 'Selma Ayşe', 'Özel', 'saozel@cu.edu.tr', 'CUUN', 'IN', TRUE),
(43, 'Professor', 'Ramazan', 'Çoban', 'rcoban@cu.edu.tr', 'CUUN', 'IN', TRUE),
(44, 'Associate Professor', 'Emine', 'Ayan', 'enacak@cu.edu.tr', 'CUUN', 'IN', TRUE),
(45, 'Associate Professor', 'Mustafa', 'Akyıldız', 'akyildizm@cu.edu.tr', 'CUUN', 'IN', TRUE),
(46, 'Professor', 'Refika', 'Altıkulaç Demirdağ', 'raltikulac@cu.edu.tr', 'CUUN', 'IN', TRUE);

INSERT INTO FACULTY (Fcode, Fname, UniCode, DeanId)
VALUES
('HAUN-ENG', 'Faculty of Engineering', 'HAUN', 21),
('YAUN-ENG', 'Faculty of Engineering', 'YAUN', 19),
('CUUN-ENG', 'Faculty of Engineering', 'CUUN', 22);

INSERT INTO DEPARTMENT (DeptCode, Name, Language, FacultyCode, ChairId)
VALUES
('YAUN-SOFT-ENG', 'Software Engineering', 'English','YAUN-ENG', 20),
('HAUN-AI-ENG', 'Artificial Intelligence Engineering', 'English','HAUN-ENG', 0),
('CUUN-COMP-ENG', 'Computer Engineering', 'English','CUUN-ENG', 24);


INSERT INTO ELECTIVE_GROUP(GroupCode,GroupName,GroupECTSTarget,GroupType)
```

VALUES

('UNV ELECT SE', 'UNIVERSITY ELECTIVE COURSE SE', 5, 'POOL_ELECT_GR'),
('ELECT SE A', 'DEPARTMEN ELECTIVE COURSE A', 6,'POOL_ELECT_GR'),
('ELECT SE B', 'DEPARTMENT ELECTIVE COURSE B', 6,'POOL_ELECT_GR'),
('AIN NO LAB','AIN Technical Elective',6,'POOL_ELECT_GR'),
('AIN WITH LAB', 'AIN Technical Elective',6,'POOL_ELECT_GR'),
('AIN LAB', 'AIN Technical Elective Lab.', 4,'POOL_ELECT_GR'),
('BBM', 'BBM Technical Elective',6,'POOL_ELECT_GR'),
('NON TEC', 'Nontechnical Elective', 4,'POOL_ELECT_GR'),
('NFE', 'Non-Field Elective Courses (University)',3,'POOL_ELECT_GR'),
('TECH ELEC 1', 'Technical Elective 1', 5,'SEM_ELECT_GR'),
('TECH ELEC 2', 'Technical Elective 2', 5,'SEM_ELECT_GR'),
('TECH ELEC 3', 'Technical Elective 3', 24,'SEM_ELECT_GR');

INSERT INTO
COURSE(Code,Name,ECTS,Theoric,Application,Laboratory,Language,DeliveryMode,LabCode,DeptCode,CourseType,Content,ElectiveType,ElectiveGroup)
VALUES

('SE 1105', 'PROGRAMMING AND PROBLEM SOLVING I', 6,3,2,0,'English', 'On-Site', NULL, 'YAUN-SOFT-ENG','Compulsory','This course is an introduction to programming course with emphasis on problem solving, programming, algorithm design and introductory programming principles', NULL, NULL),

('SE 1108', 'PROGRAMMING AND PROBLEM SOLVING II', 5,3,2,0, 'English', 'On-Site', NULL,'YAUN-SOFT-ENG', 'Compulsory', 'Classes, Methods, Inheritance, Polymorphism, Interfaces, Exceptions, Simple File Processing', NULL, NULL),

('COMP 1202','DISCRETE COMPUTATIONAL STRUCTURES', 6, 2, 2, 0, 'English', 'On-Site', NULL, 'YAUN-SOFT-ENG', 'Compulsory', 'Set theory, functions, relations, graphs, trees.', NULL, NULL),

('COMP 3330','AUTOMATA THEORY', 6, 3, 0, 0, 'English', 'On-Site', NULL, 'YAUN-SOFT-ENG', 'Compulsory', 'Finite automata, regular expressions, regular languages, pushdown automata, context-free grammars, context-free languages, Turing machines, recursive languages, recursively enumerable languages, Chomsky hierarchy, decidability.', NULL, NULL),

('SE 3317', 'SOFTWARE DESIGN AND ARCHITECTURE', 6, 2, 2, 0, 'English', 'On-Site', NULL, 'YAUN-SOFT-ENG', 'Compulsory', 'This course covers the principals behind the software design patterns and architectures design patterns, and their application in constructing software components.', NULL, NULL),

('SE 2230', 'DATABASE SYSTEMS', 6, 2, 2, 0, 'English', 'On-Site', NULL, 'YAUN-SOFT-ENG', 'Compulsory', 'This course covers relational model, entity-relationship model, normalization of relations, basic and advanced SQL using constraints, views and triggers, fine tuning of databases using indexes, and using transactions to provide ACID properties.', NULL, NULL),

('CEN111', 'Algorithms and Programming I', 4, 4, 0, 0, 'English', 'On-Site', 'CEN113', 'CUUN-COMP-ENG', 'Compulsory', 'Problem solving fundamentals. Algorithm Design. Programming Basics. Program control structures. Modular Programming', NULL, NULL),

('CEN113', 'Algorithms and Programming Lab. I', 2, 0, 2, 0, 'English', 'On-Site', NULL, 'CUUN-COMP-ENG', 'Compulsory', 'Implementation of software from CEN111 Algorithms And Programming I course', NULL, NULL),

('CEN116', 'Algorithms and Programming II', 4, 3, 0, 0, 'English', 'On-Site', 'CEN118', 'CUUN-COMP-ENG', 'Compulsory', 'Structural programming concept. C Programming Language, Fundamentals, Data types, Statements, String functions. Array manipulations. Procedures and Functions. Recursion. Sorting and Searching algorithms.', NULL, NULL),

('CEN118', 'Algorithms and Programming Lab. II', 2, 0, 2, 0, 'English', 'On-Site', NULL, 'CUUN-COMP-ENG', 'Compulsory', 'Implementation of software in the CEN116 Algorithms and Programming I course content', NULL, NULL),

('CEN343', 'Theory of Computation', 5, 3, 0, 0, 'English', 'On-Site', NULL, 'CUUN-COMP-ENG', 'Compulsory', 'Chomsky hierarchy, Regular languages, Context free languages, Turing machines, Decidability, P and NP languages, NP-complete languages', NULL, NULL),

('CEN202', 'Programming Languages', 6, 3, 0, 0, 'English', 'On-Site', NULL, 'CUUN-COMP-ENG', 'Compulsory', 'Basic programming knowledge', NULL, NULL),

('CEN301', 'Data Management and File Structures', 6, 3, 0,0, 'English', 'On-Site', NULL, 'CUUN-COMP-ENG', 'Compulsory', 'An overview of data storage and indexing methods. Disks and files. RAID systems. Disk space management. Record-based files. Tree-based directories (B+ Tree, B Tree, LSM Tree). Hash indexes (static hash, extendible hash, linear hash indexes).', NULL, NULL),

('CEN311', 'Linear Systems', 5, 3, 0, 0, 'English', 'On-Site', NULL, 'CUUN-COMP-ENG', 'Elective', 'Fourier series, Fourier transforms, transfer functions of continuous and discrete-time systems, transient and steady-state response, natural response and stability, convolution.', 'Semester Elective', 'TECH ELEC 1'),

('CEN312', 'Computer Graphics', 5, 3, 0, 0, 'English', 'On-Site', NULL, 'CUUN-COMP-ENG', 'Elective', 'Basic computer graphics algorithms are examined by using OpenGL applications. 2D and 3D graphics techniques, modelling of shapes, illumination, shading, texture mapping methods and algorithms are covered.', 'Semester Elective', 'TECH ELEC 2'),

('CEN405', 'Introduction to Robotics', 6, 3, 0,0, 'English', 'On-Site', NULL, 'CUUN-COMP-ENG', 'Elective', 'General concepts, Transformations, Advanced kinematics, Inverse kinematics, Jacobian, Orbit Planning', 'Semester Elective', 'TECH ELEC 3'),

('SD0367', 'Examinations on Modern Turkish Literature I', 3, 2, 0,0, 'English', 'On-Site', NULL, 'CUUN-COMP-ENG', 'Elective', 'The content of this course is modern Turkish poetry in the context of poetry analysis related to Tanzimat, Servet-i Funun, National Literature and Turkish poetry in the Republican period.', 'Pool Elective', 'NFE'),

('SD0499', 'Importance of Industrial Raw Materials in Our Life', 3, 2, 0,0, 'English', 'On-Site', NULL, 'CUUN-COMP-ENG', 'Elective', 'Introduction of mineral deposits formed by the effect of external events, Effects on mineral deposits in geological and chemical events, knowledge of alterations, Explanation of these mineral deposits,', 'Pool Elective', 'NFE'),

('SD0534', 'CREATIVE WRITING', 3, 2, 0, 0, 'English', 'On-Site', NULL, 'CUUN-COMP-ENG', 'Elective', 'Personal essay, academic essay, fiction, free writing, grammar, writing game, poem', 'Pool Elective', 'NFE'),

('BBM101', 'INTRODUTION TO PROGRAMMING I', 6, 3, 0, 0, 'English', 'On-Site', 'BBM103', 'HAUN-AI-ENG', 'Compulsory', 'This course is structured around fundamental topics such as control flow, functions, lists, file operations, basic data structures, function design, recursion, and memory management.', NULL, NULL),

('BBM103', 'INTRODUCTION TO PROGRAMMING LAB. I', 4, 0, 2, 0, 'English', 'On-Site', NULL, 'HAUN-AI-ENG', 'Compulsory', 'This course is structured around fundamental topics such as control flow, functions, lists, file operations, basic data structures, function design, recursion, and memory management.', NULL, NULL),

('BBM102', 'INTRODUTION TO PROGRAMMING II', 8, 3, 0, 0, 'English', 'On-Site', 'BBM104', 'HAUN-AI-ENG', 'Compulsory', 'The fundamentals of object-oriented programming.', NULL, NULL),

('BBM104', 'INTRODUCTION TO PROGRAMMING LAB. II', 4, 0, 2, 0, 'English', 'On-Site', NULL, 'HAUN-AI-ENG', 'Compulsory', 'The fundamentals of object-oriented programming.', NULL, NULL),

('EKO115', 'General Economy', 4, 3, 0, 3, 'English', NULL, NULL, 'HAUN-AI-ENG', 'Elective', NULL, 'Pool Elective', 'NON TEC'),

('AIN422', 'Deep Learning Laboratory', 4, 0, 2, 0, 'English', 'On-Site', NULL, 'HAUN-AI-ENG', 'Elective', 'Neural networks, learning in neural networks, backpropagation, supervised deep learning methods, convolutional neural networks, unsupervised deep learning methods, recurrent neural network models, fundamentals of reinforcement deep learning, modeling of sequential data, fundamentals of generative adversarial networks.', 'Pool Elective', 'AIN LAB'),

('AIN411', 'Introduction to Bioinformatics', 6, 3, 0, 0, 'English', 'On-Site', NULL, 'HAUN-AI-ENG', 'Elective', 'Fundamentals of molecular biology and genetics, and the structure and properties of data generated at different levels; DNA and protein sequence analysis; protein 3D structure analysis; gene and protein functional annotation and prediction', 'Pool Elective', 'AIN NO LAB'),

('AIN420', 'Introduction o Deep Learning', 6, 3, 0, 0, 'English', 'On-Site', 'AIN422', 'HAUN-AI-ENG', 'Elective', 'Neural networks; learning in neural networks; backpropagation; supervised deep learning methods; convolutional neural networks; unsupervised deep learning methods', 'Pool Elective', 'AIN WITH LAB'),

('BBM486', 'Design Patterns', 3, 3, 0, 0, 'English', 'On-Site', NULL, 'HAUN-AI-ENG', 'Elective', ' To teach software design patterns in an applicable way.', 'Pool Elective', 'BBM'),

('ANE 0301', 'BASIC VISUAL EFFECT I', 5, 2, 2, 0, 'English', 'On-Site', NULL, 'YAUN-SOFT-ENG', 'Elective', 'Visual effects and compositing techniques will be demonstrated from class projects.Real life video merging with green screen video and mixing with visual effects will be practiced during the lessons.', 'Pool Elective', 'UNV ELECT SE'),

('SOFL 1611', 'GERMAN I', 4, 4, 0, 0, 'English', 'On-Site', NULL, 'YAUN-SOFT-ENG', 'Elective', 'The content of this course is designed according to A1 level in the scope of Common European Framework of Reference, development of communicative and learning competences.', 'Pool Elective', 'UNV ELECT SE'),

('UFND 5020', 'ETHIC CULTURE', 2, 2, 0, 0, 'English', 'On-Site', NULL, 'YAUN-SOFT-ENG', 'Elective', 'Ethics has been devised as a product of Yaşar University's institutional scientific identity assertions. It envisages in this assertion that academic and administrative staff, students and graduates, all adopt these identity values which reflect 'an ethical consciousness of the environment'.', 'Pool Elective', 'UNV ELECT SE'),

('COMP 4312', 'THEORY OF COMPUTATION', 6, 3, 0, 0, 'English', 'On-Site', NULL, 'YAUN-SOFT-ENG', 'Elective', 'Introduction & basic definitions, concept of abstract automata, finite state machines; push-down automata, Turing Machine, Formal languages, context-free languages, the relationship between automata, languages and grammars. Computability and solvability, some applications.', 'Pool Elective', 'ELECT SE A'),

('EEE 4515', 'BIOMEDICAL IMAGING', 6, 3, 0, 0, 'English', 'On-Site', NULL, 'YAUN-SOFT-ENG', 'Elective', 'This course presents the fundamentals of medical imaging with a focus on signals and systems aspect and with up-to-date descriptions of instrumentation. Important imaging modalities are described including projection radiography, x-ray computed tomography, nuclear medicine scintigraphy and emission tomography, ultrasound imaging, and magnetic resonance imaging. Basic imaging principles with a review of two dimensional signal processing and image quality are introduced', 'Pool Elective', 'ELECT SE B'),

('AIN312', 'Formal Languages and Automata Theory', 6, 3, 0, 0, 'English', 'On-Site', NULL, 'HAUN-AI-ENG', 'Compulsory', 'Introduction to automata theory.Deterministic and nondeterministic finite automata.Regular languages and regular expressions.', NULL, NULL),

('AIN211', 'Principles of Artificial Intelligence', 6, 3, 2, 0, 'English', 'On-Site', NULL, 'HAUN-AI-ENG', 'Compulsory', 'Problem-solving agents, complete search algorithms, heuristic search algorithms, game-based algorithms, constraint satisfaction problems, logical agents, propositional logic and first-order logic, inference and decision making.', NULL, NULL);

INSERT INTO SECTION (SectionCode,SemPeriod, Year,CourseCode,CoordinatorId,AssistantId)
VALUES

('CEN111 S', 'FALL', 2025, 'CEN111', 41, NULL),
('CEN113 S', 'FALL', 2025, 'CEN113', 41, NULL),
('CEN116 S', 'SPRING', 2025, 'CEN116', 41, NULL),
('CEN118 S', 'SPRING', 2025, 'CEN118', 41, NULL),
('CEN343 S', 'FALL', 2025, 'CEN343', 24, NULL),
('CEN202 S', 'SPRING', 2025, 'CEN202', 16, NULL),
('CEN301 S', 'FALL', 2025, 'CEN301', 42, NULL),
('CEN311 S', 'FALL', 2025, 'CEN311', 43, NULL),
('CEN312 S', 'SPRING', 2025, 'CEN312', 16, NULL),
('CEN405 S', 'FALL', 2025, 'CEN405', 43, NULL),
('SD0367 S', 'FALL', 2025, 'SD0367', 44, NULL),
('SD0499 SF', 'FALL', 2025, 'SD0499', 45, NULL),
('SD0499 SS', 'SPRING', 2025, 'SD0499', 45, NULL),
('SD0534 S', 'SPRING', 2025, 'SD0534', 46, NULL),
('BBM101 S', 'FALL', 2025, 'BBM101', NULL, NULL),
('BBM103 S', 'FALL', 2025, 'BBM103', NULL, NULL),
('BBM102 S', 'FALL', 2025, 'BBM102', NULL, NULL),
('BBM104 S', 'FALL', 2025, 'BBM104', NULL, NULL),
('EKO115 SF', 'FALL', 2025, 'EKO115', NULL, NULL),
('EKO115 SS', 'SPRING', 2025, 'EKO115', NULL, NULL),
('AIN422 SF', 'FALL', 2025, 'AIN422', NULL, NULL),
('AIN422 SS', 'SPRING', 2025, 'AIN422', NULL, NULL),
('AIN411 S', 'FALL', 2025, 'AIN411', NULL, NULL),
('AIN420 SF', 'FALL', 2025, 'AIN420', NULL, NULL),
('AIN420 SS', 'SPRING', 2025, 'AIN420', NULL, NULL),
('BBM486 S', 'FALL', 2025, 'BBM486', NULL, NULL),

```
('AIN312 S', 'SPRING', 2025, 'AIN312', NULL, NULL),
('AIN211 S', 'SPRING', 2025, 'AIN211', NULL, NULL),
('SE 1105 S', 'FALL', 2025, 'SE 1105', 1, 2),
('SE 1108 S', 'SPRING', 2025, 'SE 1108', 1, NULL),
('COMP 1202 S', 'FALL', 2025, 'COMP 1202', 31, NULL),
('COMP 3330 S', 'FALL', 2025, 'COMP 3330', NULL, NULL),
('SE 3317 S', 'FALL', 2025, 'SE 3317', 34, NULL),
('SE 2230 S', 'FALL', 2025, 'SE 2230', 33, NULL),
('ANE 0301 SF', 'FALL', 2025, 'ANE 0301', 36, NULL),
('ANE 0301 SS', 'SPRING', 2025, 'ANE 0301', 36, NULL),
('SOFL 1611 SF', 'FALL', 2025, 'SOFL 1611', 37, NULL),
('SOFL 1611 SS', 'SPRING', 2025, 'SOFL 1611', 37, NULL),
('UFND 5020 SF', 'FALL', 2025, 'UFND 5020', 6, NULL),
('UFND 5020 SS', 'SPRING', 2025, 'UFND 5020', 6, NULL),
('COMP 4312 SF', 'FALL', 2025, 'COMP 4312', 40, NULL),
('COMP 4312 SS', 'SPRING', 2025, 'COMP 4312', 40, NULL),
('EEE 4515 S', 'SPRING', 2025, 'EEE 4515', 8, NULL);


INSERT INTO CURRICULUM_ITEM(DeptCode,Semester,CurriculumItemType,GroupOrCourseCode)
VALUES
('YAUN-SOFT-ENG', 1,'COMPUL', 'SE 1105'),
('YAUN-SOFT-ENG', 2,'COMPUL', 'SE 1108'),
('YAUN-SOFT-ENG', 3,'COMPUL', 'COMP 1202'),
('YAUN-SOFT-ENG', 5,'COMPUL', 'COMP 3330'),
('YAUN-SOFT-ENG', 5,'COMPUL', 'SE 3317'),
('YAUN-SOFT-ENG', 4,'COMPUL', 'SE 2230'),
('YAUN-SOFT-ENG', 5,'EL-POOL', 'UNV ELECT SE'),
('YAUN-SOFT-ENG', 8,'EL-POOL', 'UNV ELECT SE'),
('YAUN-SOFT-ENG', 6,'EL-POOL', 'ELECT SE A'),
('YAUN-SOFT-ENG', 7,'EL-POOL', 'ELECT SE A'),
('YAUN-SOFT-ENG', 7,'EL-POOL', 'ELECT SE A'),
('YAUN-SOFT-ENG', 6,'EL-POOL', 'ELECT SE A'),
('YAUN-SOFT-ENG', 8,'EL-POOL', 'ELECT SE B'),
('YAUN-SOFT-ENG', 8,'EL-POOL', 'ELECT SE B'),
('CUUN-COMP-ENG', 1,'COMPUL', 'CEN111'),
('CUUN-COMP-ENG', 1,'COMPUL', 'CEN113'),
('CUUN-COMP-ENG', 2,'COMPUL', 'CEN116'),
('CUUN-COMP-ENG', 2,'COMPUL', 'CEN118'),
('CUUN-COMP-ENG', 5,'COMPUL', 'CEN343'),
('CUUN-COMP-ENG', 4,'COMPUL', 'CEN202'),
('CUUN-COMP-ENG', 5,'COMPUL', 'CEN301'),
('CUUN-COMP-ENG', 5,'EL-SEM', 'TECH ELEC 1'),
('CUUN-COMP-ENG', 6,'EL-SEM', 'TECH ELEC 2'),
('CUUN-COMP-ENG', 7,'EL-SEM', 'TECH ELEC 3'),
('CUUN-COMP-ENG', 5,'EL-POOL', 'NFE'),
('CUUN-COMP-ENG', 6,'EL-POOL', 'NFE'),
('CUUN-COMP-ENG', 7,'EL-POOL', 'NFE'),
('CUUN-COMP-ENG', 8,'EL-POOL', 'NFE'),
('HAUN-AI-ENG', 1, 'COMPUL', 'BBM101'),
('HAUN-AI-ENG', 1, 'COMPUL', 'BBM103'),
('HAUN-AI-ENG', 2, 'COMPUL', 'BBM102'),
```

```
        ('HAUN-AI-ENG', 2, 'COMPUL', 'BBM104'),
        ('HAUN-AI-ENG', 6, 'COMPUL', 'AIN312'),
        ('HAUN-AI-ENG', 4, 'COMPUL', 'AIN211'),
        ('HAUN-AI-ENG', 4, 'EL-POOL', 'NON TEC'),
        ('HAUN-AI-ENG', 5, 'EL-POOL', 'NON TEC'),
        ('HAUN-AI-ENG', 6, 'EL-POOL', 'NON TEC'),
        ('HAUN-AI-ENG', 7, 'EL-POOL', 'NON TEC'),
        ('HAUN-AI-ENG', 8, 'EL-POOL', 'NON TEC'),
        ('HAUN-AI-ENG', 5, 'EL-POOL', 'AIN NO LAB'),
        ('HAUN-AI-ENG', 7, 'EL-POOL', 'AIN NO LAB'),
        ('HAUN-AI-ENG', 6, 'EL-POOL', 'AIN WITH LAB'),
        ('HAUN-AI-ENG', 6, 'EL-POOL', 'AIN WITH LAB'),
        ('HAUN-AI-ENG', 7, 'EL-POOL', 'AIN WITH LAB'),
        ('HAUN-AI-ENG', 8, 'EL-POOL', 'AIN WITH LAB'),
        ('HAUN-AI-ENG', 8, 'EL-POOL', 'AIN WITH LAB'),
        ('HAUN-AI-ENG', 6, 'EL-POOL', 'AIN LAB'),
        ('HAUN-AI-ENG', 6, 'EL-POOL', 'AIN LAB'),
        ('HAUN-AI-ENG', 7, 'EL-POOL', 'AIN LAB'),
        ('HAUN-AI-ENG', 8, 'EL-POOL', 'AIN LAB'),
        ('HAUN-AI-ENG', 8, 'EL-POOL', 'AIN LAB'),
        ('HAUN-AI-ENG', 5, 'EL-POOL', 'BBM');


INSERT INTO PREREQUISITE(CourseCode, PrerequisiteCode)
VALUES
        ('SE 1108', 'SE 1105'),
        ('COMP 3330', 'COMP 1202'),
        ('SE 3317', 'SE 1108'),
        ('BBM101', 'BBM103'),
        ('BBM103', 'BBM101'),
        ('BBM102', 'BBM104'),
        ('BBM102', 'BBM103'),
        ('BBM102', 'BBM101'),
        ('BBM104', 'BBM102'),
        ('BBM104', 'BBM103'),
        ('BBM104', 'BBM101'),
        ('AIN422', 'BBM102'),
        ('AIN422', 'BBM104'),
        ('AIN422', 'AIN420'),
        ('AIN411', 'BBM102'),
        ('AIN411', 'BBM104'),
        ('AIN420', 'BBM102'),
        ('AIN420', 'BBM104'),
        ('AIN420', 'AIN422'),
        ('AIN211', 'BBM101'),
        ('AIN211', 'BBM103'),
        ('BBM486', 'BBM102'),
        ('BBM486', 'BBM104');


INSERT INTO MATCHES(DeptCode, ItemCode,CourseCode)
  VALUES
```

('YAUN-SOFT-ENG',1,'CEN111'),
('YAUN-SOFT-ENG',1,'BBM101'),
('YAUN-SOFT-ENG', 2,'CEN116'),
('YAUN-SOFT-ENG', 2,'BBM102'),
('YAUN-SOFT-ENG', 4,'CEN343'),
('YAUN-SOFT-ENG', 4,'AIN312'),
('YAUN-SOFT-ENG', 6,'CEN301'),
('YAUN-SOFT-ENG', 7,'SD0367'),
('YAUN-SOFT-ENG', 7,'SD0499'),
('YAUN-SOFT-ENG', 7,'SD0534'),
('YAUN-SOFT-ENG', 7,'EKO115'),
('YAUN-SOFT-ENG', 8,'SD0367'),
('YAUN-SOFT-ENG', 8,'SD0499'),
('YAUN-SOFT-ENG', 8,'SD0534'),
('YAUN-SOFT-ENG', 8,'EKO115'),
('CUUN-COMP-ENG', 15,'BBM101'),
('CUUN-COMP-ENG', 15,'SE 1105'),
('CUUN-COMP-ENG', 17,'BBM102'),
('CUUN-COMP-ENG', 17,'SE 1108'),
('CUUN-COMP-ENG', 19,'AIN312'),
('CUUN-COMP-ENG', 19,'COMP 3330'),
('CUUN-COMP-ENG', 21,'SE 2230'),
('HAUN-AI-ENG', 29,'SE 1105'),
('HAUN-AI-ENG', 29,'CEN111'),
('HAUN-AI-ENG', 31,'SE 1108'),
('HAUN-AI-ENG', 31,'CEN116'),
('HAUN-AI-ENG', 33, 'COMP 3330'),
('HAUN-AI-ENG', 33, 'CEN343'),
('YAUN-SOFT-ENG', 9,'CEN113'),
('YAUN-SOFT-ENG', 9,'CEN118'),
('YAUN-SOFT-ENG', 9,'CEN202'),
('YAUN-SOFT-ENG', 9,'CEN301'),
('YAUN-SOFT-ENG', 9,'CEN311'),
('YAUN-SOFT-ENG', 9,'CEN312'),
('YAUN-SOFT-ENG', 9,'CEN405'),
('YAUN-SOFT-ENG', 9,'AIN422'),
('YAUN-SOFT-ENG', 9,'AIN411'),
('YAUN-SOFT-ENG', 9,'AIN420'),
('YAUN-SOFT-ENG', 9,'BBM486'),
('YAUN-SOFT-ENG', 9,'AIN211'),
('YAUN-SOFT-ENG', 10,'CEN113'),
('YAUN-SOFT-ENG', 10,'CEN118'),
('YAUN-SOFT-ENG', 10,'CEN202'),
('YAUN-SOFT-ENG', 10,'CEN301'),
('YAUN-SOFT-ENG', 10,'CEN311'),
('YAUN-SOFT-ENG', 10,'CEN312'),
('YAUN-SOFT-ENG', 10,'CEN405'),
('YAUN-SOFT-ENG', 10,'AIN422'),
('YAUN-SOFT-ENG', 10,'AIN411'),
('YAUN-SOFT-ENG', 10,'AIN420'),
('YAUN-SOFT-ENG', 10,'BBM486'),
('YAUN-SOFT-ENG', 10,'AIN211'),

('YAUN-SOFT-ENG', 11,'CEN113'),
('YAUN-SOFT-ENG', 11,'CEN118'),
('YAUN-SOFT-ENG', 11,'CEN202'),
('YAUN-SOFT-ENG', 11,'CEN301'),
('YAUN-SOFT-ENG', 11,'CEN311'),
('YAUN-SOFT-ENG', 11,'CEN312'),
('YAUN-SOFT-ENG', 11,'CEN405'),
('YAUN-SOFT-ENG', 11,'AIN422'),
('YAUN-SOFT-ENG', 11,'AIN411'),
('YAUN-SOFT-ENG', 11,'AIN420'),
('YAUN-SOFT-ENG', 11,'BBM486'),
('YAUN-SOFT-ENG', 11,'AIN211'),
('YAUN-SOFT-ENG', 12,'CEN113'),
('YAUN-SOFT-ENG', 12,'CEN118'),
('YAUN-SOFT-ENG', 12,'CEN202'),
('YAUN-SOFT-ENG', 12,'CEN301'),
('YAUN-SOFT-ENG', 12,'CEN311'),
('YAUN-SOFT-ENG', 12,'CEN312'),
('YAUN-SOFT-ENG', 12,'CEN405'),
('YAUN-SOFT-ENG', 12,'AIN422'),
('YAUN-SOFT-ENG', 12,'AIN411'),
('YAUN-SOFT-ENG', 12,'AIN420'),
('YAUN-SOFT-ENG', 12,'BBM486'),
('YAUN-SOFT-ENG', 12,'AIN211'),
('YAUN-SOFT-ENG', 13,'CEN113'),
('YAUN-SOFT-ENG', 13,'CEN118'),
('YAUN-SOFT-ENG', 13,'CEN202'),
('YAUN-SOFT-ENG', 13,'CEN301'),
('YAUN-SOFT-ENG', 13,'CEN311'),
('YAUN-SOFT-ENG', 13,'CEN312'),
('YAUN-SOFT-ENG', 13,'CEN405'),
('YAUN-SOFT-ENG', 13,'AIN422'),
('YAUN-SOFT-ENG', 13,'AIN411'),
('YAUN-SOFT-ENG', 13,'AIN420'),
('YAUN-SOFT-ENG', 13,'BBM486'),
('YAUN-SOFT-ENG', 13,'AIN211'),
('YAUN-SOFT-ENG', 14,'CEN113'),
('YAUN-SOFT-ENG', 14,'CEN118'),
('YAUN-SOFT-ENG', 14,'CEN202'),
('YAUN-SOFT-ENG', 14,'CEN301'),
('YAUN-SOFT-ENG', 14,'CEN311'),
('YAUN-SOFT-ENG', 14,'CEN312'),
('YAUN-SOFT-ENG', 14,'CEN405'),
('YAUN-SOFT-ENG', 14,'AIN422'),
('YAUN-SOFT-ENG', 14,'AIN411'),
('YAUN-SOFT-ENG', 14,'AIN420'),
('YAUN-SOFT-ENG', 14,'BBM486'),
('YAUN-SOFT-ENG', 14,'AIN211'),

('CUUN-COMP-ENG', 22,'AIN422'),
('CUUN-COMP-ENG', 22,'AIN411'),

```
('CUUN-COMP-ENG', 22,'AIN420'),
('CUUN-COMP-ENG', 22,'BBM486'),
('CUUN-COMP-ENG', 22,'AIN211'),
('CUUN-COMP-ENG', 22,'COMP 1202'),
('CUUN-COMP-ENG', 22,'SE 3317'),
('CUUN-COMP-ENG', 22,'COMP 4312'),
('CUUN-COMP-ENG', 22,'EEE 4515'),
('CUUN-COMP-ENG', 23,'AIN422'),
('CUUN-COMP-ENG', 23,'AIN411'),
('CUUN-COMP-ENG', 23,'AIN420'),
('CUUN-COMP-ENG', 23,'BBM486'),
('CUUN-COMP-ENG', 23,'AIN211'),
('CUUN-COMP-ENG', 23,'COMP 1202'),
('CUUN-COMP-ENG', 23,'SE 3317'),
('CUUN-COMP-ENG', 23,'COMP 4312'),
('CUUN-COMP-ENG', 23,'EEE 4515'),
('CUUN-COMP-ENG', 24,'AIN422'),
('CUUN-COMP-ENG', 24,'AIN411'),
('CUUN-COMP-ENG', 24,'AIN420'),
('CUUN-COMP-ENG', 24,'BBM486'),
('CUUN-COMP-ENG', 24,'AIN211'),
('CUUN-COMP-ENG', 24,'COMP 1202'),
('CUUN-COMP-ENG', 24,'SE 3317'),
('CUUN-COMP-ENG', 24,'COMP 4312'),
('CUUN-COMP-ENG', 24,'EEE 4515'),
('CUUN-COMP-ENG', 25,'ANE 0301'),
('CUUN-COMP-ENG', 25,'SOFL 1611'),
('CUUN-COMP-ENG', 25,'UFND 5020'),
('CUUN-COMP-ENG', 25,'EKO115'),
('CUUN-COMP-ENG', 26,'ANE 0301'),
('CUUN-COMP-ENG', 26,'SOFL 1611'),
('CUUN-COMP-ENG', 26,'UFND 5020'),
('CUUN-COMP-ENG', 26,'EKO115'),
('CUUN-COMP-ENG', 27,'ANE 0301'),
('CUUN-COMP-ENG', 27,'SOFL 1611'),
('CUUN-COMP-ENG', 27,'UFND 5020'),
('CUUN-COMP-ENG', 27,'EKO115'),
('CUUN-COMP-ENG', 28,'ANE 0301'),
('CUUN-COMP-ENG', 28,'SOFL 1611'),
('CUUN-COMP-ENG', 28,'UFND 5020'),
('CUUN-COMP-ENG', 28,'EKO115'),

('HAUN-AI-ENG', 35,'ANE 0301'),
('HAUN-AI-ENG', 35,'SOFL 1611'),
('HAUN-AI-ENG', 35,'UFND 5020'),
('HAUN-AI-ENG', 35,'SD0367'),
('HAUN-AI-ENG', 35,'SD0499'),
('HAUN-AI-ENG', 35,'SD0534'),
('HAUN-AI-ENG', 36,'ANE 0301'),
('HAUN-AI-ENG', 36,'SOFL 1611'),
('HAUN-AI-ENG', 36,'UFND 5020'),
('HAUN-AI-ENG', 36,'SD0367'),
```

```
	('HAUN-AI-ENG', 36,'SD0499'),
	('HAUN-AI-ENG', 36,'SD0534'),
	('HAUN-AI-ENG', 37,'ANE 0301'),
	('HAUN-AI-ENG', 37,'SOFL 1611'),
	('HAUN-AI-ENG', 37,'UFND 5020'),
	('HAUN-AI-ENG', 37,'SD0367'),
	('HAUN-AI-ENG', 37,'SD0499'),
	('HAUN-AI-ENG', 37,'SD0534'),
	('HAUN-AI-ENG', 38,'ANE 0301'),
	('HAUN-AI-ENG', 38,'SOFL 1611'),
	('HAUN-AI-ENG', 38,'UFND 5020'),
	('HAUN-AI-ENG', 38,'SD0367'),
	('HAUN-AI-ENG', 38,'SD0499'),
	('HAUN-AI-ENG', 38,'SD0534'),
	('HAUN-AI-ENG', 39,'ANE 0301'),
	('HAUN-AI-ENG', 39,'SOFL 1611'),
	('HAUN-AI-ENG', 39,'UFND 5020'),
	('HAUN-AI-ENG', 39,'SD0367'),
	('HAUN-AI-ENG', 39,'SD0499'),
	('HAUN-AI-ENG', 39,'SD0534'),
	('HAUN-AI-ENG', 40,'EEE 4515'),
	('HAUN-AI-ENG', 40,'CEN405'),
	('HAUN-AI-ENG', 41,'EEE 4515'),
	('HAUN-AI-ENG', 41,'CEN405'),
('HAUN-AI-ENG', 52,'CEN312');
```

6.4 Triggers

```
-- When adding or changing assistant and coordinator in a SECTION, a check is performed to see if there are
already assistant and coordinator.
CREATE OR REPLACE FUNCTION trg_section_validate_members()
RETURNS TRIGGER AS $$
BEGIN
	-- Check Coordinator
	IF NEW.CoordinatorId IS NOT NULL THEN
	IF NOT EXISTS (
	SELECT 1
	FROM FACULTY_MEMBER
	WHERE MemberId = NEW.CoordinatorId
	AND MemberType = 'IN'
	AND IsCoordinator = TRUE
	) THEN
	RAISE EXCEPTION 'CoordinatorId % is not a Coordinator', NEW.CoordinatorId;
	END IF;
	END IF;

	-- Check Assistant
	IF NEW.AssistantId IS NOT NULL THEN
	IF NOT EXISTS (
	SELECT 1
	FROM FACULTY_MEMBER
	WHERE MemberId = NEW.AssistantId
	AND MemberType = 'AS'
	) THEN
```

```
		RAISE EXCEPTION 'AssistantId % is not an Assistant', NEW.AssistantId;
		END IF;
		END IF;

		RETURN NEW;
END;
$$ LANGUAGE plpgsql;

CREATE TRIGGER trg_section_member_check
BEFORE INSERT OR UPDATE ON SECTION
FOR EACH ROW
EXECUTE FUNCTION trg_section_validate_members();


-- When an item is added to CURRICULUM_ITEM, it should be automatically added to the appropriate sub-
table.
CREATE OR REPLACE FUNCTION trg_curriculum_item_router()
RETURNS TRIGGER AS $$
BEGIN
		-- Pool elective
		IF NEW.CurriculumItemType = 'EL-POOL'
		AND (
			SELECT GroupType
			FROM ELECTIVE_GROUP
			WHERE GroupCode = NEW.GroupOrCourseCode
		)  ='POOL_ELECT_GR'
			THEN
		INSERT INTO ELECTIVE_POOL_ITEM
		(DeptCode, ItemCode, ElectiveCode)
		VALUES
		(NEW.DeptCode, NEW.ItemCode, NEW.GroupOrCourseCode);

		-- Semester elective
		ELSIF NEW.CurriculumItemType = 'EL-SEM'
		AND (
			SELECT GroupType
			FROM ELECTIVE_GROUP
			WHERE GroupCode = NEW.GroupOrCourseCode
		) ='SEM_ELECT_GR'
		THEN
		INSERT INTO ELECTIVE_SEMESTER_ITEM
		(DeptCode, ItemCode, ElectiveCode)
		VALUES
		(NEW.DeptCode, NEW.ItemCode, NEW.GroupOrCourseCode);

		-- Compulsory
		ELSE
		INSERT INTO COMPULSORY_ITEM
		(DeptCode, ItemCode, CompulsoryCode)
		VALUES
		(NEW.DeptCode, NEW.ItemCode, NEW.GroupOrCourseCode);
		END IF;

		RETURN NEW;
END;
$$ LANGUAGE plpgsql;

CREATE TRIGGER curriculum_item_after_insert
AFTER INSERT ON CURRICULUM_ITEM
FOR EACH ROW
```

```
EXECUTE FUNCTION trg_curriculum_item_router();


-- Prevent direct data insertion into sub-item tables.
CREATE OR REPLACE FUNCTION block_item_manual_insert()
RETURNS TRIGGER AS $$
BEGIN
	-- Allow if the trigger is called by another trigger.
	IF pg_trigger_depth() > 0 THEN
	RETURN NEW;
	END IF;

	-- Block the user if they try to add it manually.
	RAISE EXCEPTION
	'Do not insert into this item table directly. Use CURRICULUM_ITEM.';
END;
$$ LANGUAGE plpgsql;

CREATE TRIGGER trg_block_pool_item_insert
BEFORE INSERT ON ELECTIVE_POOL_ITEM
FOR EACH ROW
EXECUTE FUNCTION block_item_manual_insert();

CREATE TRIGGER trg_block_semester_item_insert
BEFORE INSERT ON ELECTIVE_SEMESTER_ITEM
FOR EACH ROW
EXECUTE FUNCTION block_item_manual_insert();

CREATE TRIGGER trg_block_compulsory_item_insert
BEFORE INSERT ON COMPULSORY_ITEM
FOR EACH ROW
EXECUTE FUNCTION block_item_manual_insert();

CREATE OR REPLACE FUNCTION trg_course_elective_group_type_check()
RETURNS TRIGGER AS $$
BEGIN
	----Pool elective
	IF NEW.CourseType = 'Elective'
	AND NEW.ElectiveType = 'Pool Elective'
	AND NOT (
	SELECT GroupType
	FROM ELECTIVE_GROUP
	WHERE GroupCode = NEW.ElectiveGroup
	) = 'POOL_ELECT_GR'
	THEN
	RAISE EXCEPTION
	'Pool Elective courses must be linked to a POOL_ELECT_GR elective group.';
	END IF;

	----Semester Elective
	IF NEW.CourseType = 'Elective'
	AND NEW.ElectiveType = 'Semester Elective'
	AND NOT (
	SELECT GroupType
	FROM ELECTIVE_GROUP
	WHERE GroupCode = NEW.ElectiveGroup
	) = 'SEM_ELECT_GR'
	THEN
	RAISE EXCEPTION
	'Semester Elective courses must be linked to a SEM_ELECT_GR elective group.';
```

```
        END IF;

        RETURN NEW;
END;
$$ LANGUAGE plpgsql;

CREATE TRIGGER trg_course_elective_group_type_check
BEFORE INSERT OR UPDATE
ON COURSE
FOR EACH ROW
EXECUTE FUNCTION trg_course_elective_group_type_check();
```

6.5 Check Constraints for Curriculum and Equivalency Rules

chk_member_type from FACULTY_MEMBER : It limits the values that MemberType can take.

chk_assistant_coordinator from FACULTY_MEMBER : if member is Assistant, then Coordinator must be FALSE

pk_member_education from MEMBER_EDUCATION : A multivalue primary key is determined.

check_elective_group_type from ELECTIVE_GROUP : It limits the values that GroupType can take.

chk_elective_type from COURSE : It limits the values that ElectiveType can take.

chk_course_type from COURSE : It limits the values that CourseType can take.

chk_course_total_hours from COURSE :The sum T+A+L cannot be less than 0.

chk_course_elective_logic from COURSE : If ElectiveType = 'Pool Elective', then ElectiveGroup can be any group in the ELECTIVE_GROUP table
If ElectiveType = 'Semester Elective', then ElectiveGroup must be NULL
If it's Compulsory, then ElectiveType = NULL and ElectiveGroup = NULL

chk_sem_period from SECTION : It limits the values that SemPeriod can take.

chk_semester from CURRICULUM_ITEM : It limits the values that Semester can take.

chk_curriculum_item_type from CURRICULUM_ITEM : It limits the values that CurriculumItemType can take.

pk_prerequisite from PREREQUISITE : A multi valuable primary key is determined.

pk_subjects from SUBJECTS : A multi valuable primary key is determined.

6.6 Data Manipulation Examples (INSERT / UPDATE / DELETE)

```
INSERT INTO FACULTY_MEMBER
VALUES (
   60,
   NULL,
   'Ahmet',
   'Ayça',
   NULL,
   'HAUN',
   'IN',
   FALSE
);

INSERT INTO TEACHES
VALUES (
        'EKO115 SF',
        60
);
```

```
INSERT INTO ELECTIVE_GROUP
VALUES(
        'TECH ELEC 4',
        'TEchnical Elective 4',
        24,
        'SEM_ELECT_GR'
);

UPDATE COURSE
SET ECTS = 6,
   DeliveryMode = 'On-Site'
WHERE Code = 'CEN343';

UPDATE TEACHES
SET InstructorId = 5
WHERE SectionCode = 'AIN411 S';

UPDATE MEMBER_EDUCATION
SET University = 'Hacettepe University',
        Field = 'Computer Engineering'
WHERE MemberId = 16 AND DegreeType = 'Master';

DELETE FROM LEARNING_OUTCOMES
WHERE CourseCode = 'CEN113' AND OutcomeId = 5 ;

DELETE FROM SECTION
WHERE SectionCode = 'CEN113 S';

DELETE FROM PREREQUISITE
WHERE CourseCode = 'BBM103'
```

6.7 Query Set: Core SELECT Statements

Query 1.

```
CREATE VIEW YAUN_curriculum AS
SELECT
        ci.ItemCode,
        ci.DeptCode,
        ci.Semester,
        ci.CurriculumItemType,
        ci.GroupOrCourseCode AS Code,
        CASE
        WHEN ci.CurriculumItemType = 'COMPUL' THEN c.Name
        WHEN ci.CurriculumItemType = 'EL-SEM' THEN c.Name
        WHEN ci.CurriculumItemType = 'EL-POOL' THEN eg.GroupName
        ELSE 'Unknown'
        END AS ItemName,
        CASE
        WHEN ci.CurriculumItemType = 'COMPUL' THEN c.ECTS
        WHEN ci.CurriculumItemType = 'EL-SEM' THEN eg.GroupECTSTarget
        WHEN ci.CurriculumItemType = 'EL-POOL' THEN eg.GroupECTSTarget
        END AS ECTS
FROM
```

```
        CURRICULUM_ITEM ci
LEFT JOIN
        COURSE c ON ci.GroupOrCourseCode = c.Code
LEFT JOIN
        ELECTIVE_GROUP eg ON ci.GroupOrCourseCode = eg.GroupCode
ORDER BY
        ci.DeptCode, ci.Semester;
```

Query 2.

```
SELECT
        ci.DeptCode,
        ci.Semester,
        ci.CurriculumItemType,
        ci.Code AS TargetCode,
        ci.ItemName AS TargetName,
        ci.ECTS AS TargetECTS,
        m.CourseCode as MatchCourseCode,
        c.Name AS MatchName,
        c.ECTS AS MatchECTS
FROM MATCHES AS m
RIGHT JOIN YAUN_curriculum AS ci
ON ci.ItemCode = m.ItemCode
        AND ci.DeptCode = m.DeptCode
JOIN COURSE AS c
ON m.CourseCode = c.Code
WHERE ci.DeptCode = 'YAUN-SOFT-ENG' AND ci.Semester = 1;
```

Query 3.

```
SELECT
        ci.Semester,
        ci.CurriculumItemType,
        COALESCE(c.Name, eg.GroupName) AS ItemName,
        COALESCE(c.ECTS, eg.GroupECTSTarget) AS ECTS,
        c.Theoric,
        c.Application,
        c.Laboratory,
        c.Content
FROM CURRICULUM_ITEM ci
LEFT JOIN COURSE c
        ON ci.GroupOrCourseCode = c.Code
LEFT JOIN ELECTIVE_GROUP eg
        ON ci.GroupOrCourseCode = eg.GroupCode
WHERE ci.DeptCode = 'HAUN-AI-ENG'
ORDER BY ci.Semester;
```

Query 4.

```
SELECT FM.LName, ME.DegreeType, ME.University, ME.Field
```

```
FROM SECTION AS S,
        TEACHES AS T,
        MEMBER_EDUCATION AS ME,
        FACULTY_MEMBER as FM
WHERE S.CourseCode = 'SE 1105'
        AND S.SectionCode = T.SectionCode
        AND T.InstructorId = ME.MemberId
        AND ME.MemberId = FM.MemberId;
```

Query 5.

```
SELECT DISTINCT ci.GroupOrCourseCode, c.Name, c.ECTS, c.Content
FROM CURRICULUM_ITEM AS ci
JOIN COURSE AS c
ON ci.GroupOrCourseCode = c.ElectiveGroup
WHERE ci.CurriculumItemType <> 'COMPUL' ;
```

6.8 Academic Insight Queries (Original SELECT Statements)

```
-- Analysis of the Balance Between Elective and Compulsory Courses in Departments. How flexible is
the department?
SELECT
   d.DeptCode,
   d.Name AS DepartmentName,
   SUM(CASE WHEN ci.CurriculumItemType = 'COMPUL' THEN 1 ELSE 0 END) AS
CompulsoryCount,
   SUM(CASE WHEN ci.CurriculumItemType LIKE 'EL-%' THEN 1 ELSE 0 END) AS
ElectiveCount
FROM CURRICULUM_ITEM AS ci
JOIN DEPARTMENT AS d
   ON d.DeptCode = ci.DeptCode
GROUP BY d.DeptCode, d.Name;


-- Prerequisite chain. Which subjects put students in a vicious cycle?
SELECT
   c.Code AS CourseCode,
   c.Name,
   COUNT(p.PrerequisiteCode) AS PrerequisiteCount
FROM COURSE AS c
LEFT JOIN PREREQUISITE AS p
   ON c.Code = p.CourseCode
GROUP BY c.Code, c.Name
ORDER BY PrerequisiteCount DESC;


-- Courses that are compulsory but have no prerequisites
SELECT DISTINCT
   c.Code,
   c.Name
FROM COURSE AS c
JOIN PREREQUISITE AS p
   ON p.CourseCode = c.Code
WHERE c.CourseType = 'Elective';
```

```
-- Courses lacking weekly content. Curriculum quality is reviewed.
SELECT
   c.Code,
   c.Name,
   COUNT(s.Week) AS DefinedWeeks
FROM COURSE AS c
LEFT JOIN SUBJECTS AS s
   ON s.CourseCode = c.Code
GROUP BY c.Code, c.Name
HAVING COUNT(s.Week) < 14;


-- Courses without resources. Curriculum quality is reviewed
SELECT
   c.Code,
   c.Name
FROM COURSE AS c
LEFT JOIN RESOURCES AS r
   ON r.CourseCode = c.Code
WHERE r.ResourceId IS NULL;
```

7. References

Batini, C., & Lenzerini, M. (1984). A methodology for data schema integration in the Entity–Relationship model. IEEE Transactions on Software Engineering, 10(6), 650–664.

Batini, C., Lenzerini, M., & Navathe, S. B. (1986). A comparative analysis of methodologies for database schema integration. ACM Computing Surveys, 18(4), 323–364.

Elmasri, R., & Navathe, S. B. (2016). Fundamentals of Database Systems (7th ed.). Pearson Education.

Course Lecture Notes. (2025). Introduction to Databases. Ege University, Faculty of Engineering, Computer Engineering Department.

Artificial Intelligence–Based Tools. (2025). Used for language refinement and consistency checking during report preparation.